MIL-W-25140B
31 October 1977
SUPERSEDING
MIL-W-25140A
19 April 1973

MILITARY SPECIFICATION

WEIGHT AND BALANCE CONTROL SYSTEM (FOR AIRCRAFT AND ROTORCRAFT)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for Weight and Balance Control Systems to provide for the management of mass characteristics programs and to facilitate the preparation and submission of data in standard format during the life of an aircraft or rotorcraft contract (see 3.3).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-F-5572	Fuel, Aircraft Reciprocating Engine, Grades 80, 91/96, 100/130, 115/145
MIL-F-5624	Fuel, Aircraft Turbine and Jet Engine Grades JP-3, JP-4, and JP-5
MIL-M-5920	Manuals, Technical, Basic Weight Check List and Loading Data

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Engineering Specifications and Standards Department (Code 93), Naval Air Engineering Center, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1500

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SPECIFICATIONS (Continued)

APPROVED WEIGHING FORMS

AN-9249	Aircraft Actual Weight and Horizontal Balance
AN-9250	Aircraft Actual Weight and Vertical Balance
DD Form 365B	Aircraft Weighing Record

STANDARDS

MILITARY

MIL-STD-480	Configuration Control - Engineering Changes, Deviations and Waivers
MIL-STD-1374	Weight and Balance Data Reporting Forms for Aircraft (Including Rotorcraft)
Part I	Group Weight Statement
Part II	Detail Weight Statement
Part III	Weight and Balance Status

PUBLICATIONS

TECHNICAL MANUALS

The requirements for these manuals are specified in the text of this specification under the title "Weight Handbook." Reference to any one of the service manuals in the Contract Data Requirements List (DD Form 1423) shall mean "Weight Handbook."

T.O. 1-1B-40 (Air Force)	Handbook, Weight and Balance Data
T.O. 1-1B-50 (Air Force)	
NA 01-1B-40 (Navy)	
TM 55-405-9 (Army)	

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General. The data covered by this specification are intended for use in determining compliance with contractual requirements for the management of weight and balance control systems of aircraft during design and construction, in establishing a system of weight and balance control for operation of the aircraft, and in preparing balance computers for class 2 aircraft. The instructions and requirements of MIL-STD-1374 shall be applied in the compilation of data for Parts I and II, the Group and Detail Weight Statements, required in aircraft weight reports.

3.2 Weight and balance control system. The contractor is required to provide a Weight and Balance Control System that will ensure Weight Control Management (1) through assignment of competent personnel, (2) by utilizing adequate measuring devices and facilities, (3) by defining techniques to be initiated to offset adverse trends, (4) by defining requirements to sub-contractors and (5) by holding constructive periodic design reviews. The contractor's system is subject to review and validation by the procuring agency. During pre-award negotiation or immediately following contract award, representatives of the prime contractor or associate contractors shall consult with the procuring agency in certifying the procedure for Program Weight and Balance Control including the submission of a Weight Control and Management Program Plan Report. Part of the Plan Report shall be devoted to specifying the detailed interface between the Work Breakdown Structure system as described by MIL-STD-881A and the Group and Detailed Weight Statements.

3.3 Submittal of data. The weight and balance data specified herein shall be submitted by the contractor during study, research and development or production contracts in accordance with the data acquisition provisions of the contract. Normally, Contract Data Requirements List (DD Form 1423) identifies by Data Item Description (DD Form 1664) the quantity, frequency and distribution of specific weight and balance data for a contract. For other contracts, data may be specified by reference to specific paragraphs herein.

3.3.1 Specification weight derivation. During the negotiation stage of preparing a contract detail specification, the contractor shall furnish a derivation of the vehicle weight for each draft of the specification until the weight status of the final draft is represented. The derivation shall be based on the design proposal (on prototype or actual data when available) and shall be furnished on the Status Report Form MIL-STD-1374, Part III. A description shall be appended explaining the reasons for the weight changes since the previous draft. A list of weight of Government Furnished Equipment, upon which the weights are based, shall be attached.

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3.3.2 Post design weight analysis report. This report shall be prepared for an aircraft designated by the procuring agency and shall be submitted no later than 90 days after the aircraft is weighed. The purpose of the report is to provide, in a single document, a record of design criteria, integrated design concept, and weight information which will serve as a baseline working document to be used during the acquisition phase and contract service life. A chronology of major design and/or weight changes will be included to provide background information on the aircraft design evolution. The detailed content and format of this report shall be prepared by the contractor and submitted for approval to the procuring agency. Compilation of the data for this report shall begin as of the contract date. It shall be updated commensurate with the design development of the aircraft and shall be maintained in an informal but organized status. This informal data shall be available for review by the procuring agency as an aid during program weight reviews. The report shall be divided into four major sections consisting of:

a. Pre-contract development. This section shall document the configuration development from the contractor's initial proposal and ending with the aircraft configuration at contract award.

b. Contract design development. This section shall document the mass properties control program, including a history of weight and design changes resulting from this program. This section shall also include all significant design development changes, in chronological order, from contract award to the designated aircraft.

c. Detail design analysis. This section shall present the description and analysis of the designated aircraft. Included in this section shall be an aircraft general description, Group and Detail Weight Statements, criteria, and detail design description and functional analyses for each structural and system group.

d. Appendices. The procuring agency will require, unless specifically waived, an appendix to this report. The agency will specify the effectivity of the appendix (i.e. to which aircraft, by production number, the appendix shall be applicable) and the submittal data, which will usually be close to the IOC (initial operational capability) date.

(1) The appendix shall describe the changes made to the given aircraft as well as a short description of the service problems or defects that precipitated the changes. It is intended that the appendix to the Post Design Weight Analysis Report will provide a running account of the weight history from the designated aircraft of paragraph c above.

(2) On subsequent contracts the procuring agency may require further appendices to cover later changes. In such cases the procuring agency will specify the submittal dates and effectivities as above.

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3.4 Weight and balance classification. For weight and balance control purposes, aircraft are divided into the following classifications.

3.4.1 Class 1A. Class 1A aircraft are those whose recommended weight or c.g. limits can not be exceeded by loading arrangements normally employed in tactical operations, and therefore need no loading control.

3.4.2 Class 1B. Class 1B aircraft are those whose recommended weight or c.g. limits sometimes can be exceeded by loading arrangements normally employed in tactical operations, and therefore need loading control.

3.4.3 Class 2. Class 2 aircraft are those whose recommended weight or c.g. limits can readily be exceeded by loading arrangements normally employed in tactical operations, and therefore need a high degree of loading control.

3.4.4 Classification assignment. Weight and balance classifications will be determined by the procuring activity and assigned in the aircraft detail specification.

3.5 Handbooks and balance computers. The requirements outlined below for the Weight Handbook and balance computer apply to production and preproduction aircraft.

3.5.1 Class 1A aircraft. There are no requirements for data to be delivered aboard Class 1A aircraft. Charts A and E for the Weight Handbook as specified herein shall be furnished to the procuring activity to provide data for use by field weight and balance personnel and overhaul activities in periodic weight and balance checks.

3.5.2 Class 1B aircraft. The contractor shall complete the Weight Handbook which shall be placed in the data case of each aircraft prior to delivery.

3.5.3 Class 2 aircraft. The contractor shall complete the Weight Handbook which shall be placed in the data case of each aircraft prior to delivery. If Government-furnished balance computers are not received prior to delivery of any aircraft on Navy contracts, the information furnished in Figure 1 shall be inserted in the front of the applicable Weight Handbooks.

3.6 Weight and balance data.

3.6.1 Preproduction contract. Unless otherwise specified, the following weight and balance data, and necessary weighings, are required under a preproduction contract:

a. Estimated Weight Report.

b. Weight and Balance Status Report.

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IMPORTANT NOTICE

Subj: Model _____, Balance Computer - Instructions Concerning

The balance computer, Load Adjuster Plate _____, was not available for installation in this aircraft prior to delivery. Chart E shall be used in the interim. The weight and balance personnel of the activity receiving this aircraft shall complete the form shown below and forward it to the indicated addressee. Arrangements will be made to have the Load Adjuster furnished as soon as available. Upon receipt of the Load Adjuster, the following shall be accomplished:

(1) Install the Load Adjuster on the clip located _____(Contractor shall give location).

(2) Insert the Instruction Pamphlet (provided with each Load Adjuster) at the end of Section 3 of the Weight Handbook installed in the aircraft.

(3) Remove this sheet from the handbook.

To: Commander, Naval Air Systems Command
Attn: Evaluation Division, Weight Control Branch (AIR-5062)
Washington, D.C. 20361

Model _____ Aircraft, Bureau No. _____, was delivered by the contractor prior to the availability of the Load Adjuster. Arrange for shipment of the one Load Adjuster, Plate _____, to the following addressee.

_____

_____

_____

_____

(Contractor to fill in aircraft model, and plate number if available)

Signature _____
(Weight and Balance Personnel)

FIGURE 1. Notice to be inserted in the Weight Handbook if Government-furnished balance computers are not received prior to aircraft delivery.

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c. Calculated Weight Report: The requirements for the Calculated Weight Report may be eliminated or reduced upon specific approval by the procuring activity.

d. Sample Charts A and E.

e. Actual Weight Report - First Aircraft.

f. Appendices to Actual Weight Reports required prior to conducting procuring activity tests.

g. Final Appendices to Actual Weight Reports.

h. Actual Weight Report: Intermediate and Last Aircraft. If a contract provides for more than one preproduction aircraft, an Actual Weight Report - Intermediate Aircraft shall be submitted for each additional aircraft unless otherwise stated in the contract. If only minor changes have been made since the last previous Actual Weight Report, a report detailing these changes, (including their effect on actual and contract/guarantee weight and balance, and the contractor responsibility under or overweight) will be sufficient.

i. If alternate power plant installations, alternate wings, or other alternate major components are included in the program, weight and balance information shall be supplied for all alternate items and for the complete aircraft with the alternate items installed. These data shall be included as an appendix to the applicable calculated and actual weight reports. Actual weighings of alternate components and, if necessary, the complete aircraft with alternate components installed shall be made to insure accurate data.

3.6.2 Production contract. The following weight and balance data, and necessary weighings, are required under a production contract:

a. Estimated Weight Report.

b. Weight and Balance Status Report: Required when specifically requested by the procuring activity.

c. Calculated Weight Report: If an Actual Weight Report has been submitted for preproduction aircraft and only minor changes have been made on the production aircraft, a report detailing these changes and their effect on the calculated and contract/guaranteed weight and balance will be sufficient.

d. Sample Charts A and E.

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e. Actual Weight Report - First Aircraft: With the specific approval of the procuring activity, the Detail Weight Statement (MIL-STD-1374, Part II), the weight empty detail balance calculations, and all vertical c.g. data may be omitted from this report, and furnished in the report for the fifth aircraft.

f. Appendices to Actual Weight Reports required prior to conducting procuring activity tests.

g. Final Appendices to Actual Weight Reports.

h. Actual Weight Report - Intermediate Aircraft: Reports shall be submitted as follows:

(1) For the 5th and 10th aircraft.

(2) Each 20th aircraft. Weighings or reports may also be required for any other aircraft when specifically requested by the procuring activity. This applies particularly in the case of production contracts for a small quantity of aircraft, or a very low production rate. Where possible, reports shall be scheduled for the first aircraft in which major changes or an accumulation of minor changes are incorporated. Therefore, the contractor shall submit to the procuring activity a proposed revised schedule when the production blocks are initially established, when major changes, program deceleration, cumulative effect of changes and major rescheduling of blocks warrant further adjustment, and when selection of aircraft for procuring activity tests (Navy demonstration and trials) is made. Minor deviations from the schedule may be approved by the local representative of the procuring activity, and appropriate note of the deviation included in the letter submitting the report.

i. Actual Weight Report - Last Aircraft.

j. If a contract is for the same type of aircraft, without major changes, and is to run consecutively to a previous contract, any or all of the following modifications to the above requirements may be made, subject to specific approval by the procuring activity:

(1) Omit the Estimated, Calculated, and Weight Status Reports.

(2) Omit the Actual Weight Report - Last Aircraft on the previous contract.

(3) Substitute an Actual Weight Report - Intermediate Aircraft for the Actual Weight Report - First Aircraft.

(4) Omit the appendices to the Actual Weight Reports.

(5) Omit the Actual Weight Report - Intermediate Aircraft for the 5th and 10th aircraft.

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k. Printed copies of released and revised Charts A and E as specified in 3.7.9.6.

3.6.3 Engineering Change Proposals (ECP's). Data affecting weight and balance shall be furnished in the comparable form of Figure 2 as an enclosure to appropriate ECP's. The submittal and preparation of ECP's is governed by MIL-STD-480, Configuration Control - Engineering Changes, Deviations and Waivers. A separate copy of the weight breakdown of the ECP shall be submitted directly to the cognizant Air Force mass properties engineer when specifically requested. The data shall be resubmitted, as above, when ECP's are revised or when Specification Change Notices (SCN's) and Contract Change Notices (CCN's) differ from previously submitted data. Summaries may be submitted in lieu of the Figure 2 format when the resubmittal is identical to the initial ECP weight data. The following requirements also apply:

a. The weight breakdown shall be furnished in enough detail to allow weight accountability for MIL-STD-1374, Part II items affected. No breakdown is required for a structural change of less than 10 pounds when the total ECP affects only one air vehicle part. Change affecting more than one part shall be treated in a manner indicated by Figure 2 regardless of the net weight change involved. The weight report on which the weight breakdown and balance data are based shall be referenced by number, date, and title. Weights of Government-Furnished Aircraft Equipment (GFAE) items deleted shall be the specified weights on which the current guaranteed weight empty is based. The weights of GFAE items added, upon which the revised contract/guaranteed weight empty will be based, shall specifically be listed. In any case, where the change in contract/guaranteed weight empty will differ materially from the change in predicted weight empty, because of differences between specified and predicted weights of GFAE or other reasons, appropriate comment shall be included calling attention to and indicating the magnitude of the differences in weight and c.g. changes.

b. The useful load change shall be given for all missions on which performance guarantees are based and for all weight conditions that are critical for design of the vehicle.

c. When a proposed change involves a large c.g. shift, information or comment shall be included to justify that loadings exceeding anticipated c.g. limits will not result.

d. When a proposed change involves a major modification of one aircraft or group of aircraft, the data required above shall be submitted in the form of a brief weight report deriving the calculated and contract/guaranteed weight and balance of the modified aircraft, normal loading, and the most forward and aft loading conditions.

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ECP NO. _____
Date _____
Prepared by: _____

MODEL XYZ - CONTRACT 00000 - WEIGHT INCREASE
FOR CHANGE FROM TWO TO FOUR 20MM GUN INSTALLATION

	GFAE	CONTRACTOR-FURNISHED MATERIAL	ARM INCHES	MOMENT INCH. LBS.
WEIGHT EMPTY				
ITEMS REMOVED	(0)	(- 81.5)		(-22675)
Gun and Ammunition Box Access Doors		- 19.0	275.0	- 5225
Ammunition Feed and Case Ejection Chutes		- 1.3	273.5	- 356
Gun Mounts and Supports		- 4.2	254.8	- 1070
Cabling, Wiring and Plugs		- 17.0	307.3	- 5224
Gun Beams and Support Frames		- 34.0	270.0	- 9180
Ammunition Compartment Floor		- 6.0	270.0	- 1620
ITEMS ADDED	(0)	(+122.0)		(+33762)
Gun and Ammunition Box Access Doors		28.5	274.5	7823
Ammunition Feed and Case Ejection Chutes		3.6	273.5	985
Gun Mounts and Supports		8.4	254.8	2140
Cabling, Wiring and Plugs		21.7	307.3	6668
Gun Beams and Support Frames		49.8	270.0	13446
Ammunition Compartment Floor		10.0	270.0	2700
	0	40.5		
Total Weight Empty Change	+ 40.5 lb.			+11087
USEFUL LOAD				
ITEMS ADDED:				
Gun, Adapter & Retainers (2)	204.8		269.0	55091
Aft Gun Mount - UPR. (2)		4.0	272.0	1088
Feed Mechanism - UPR. L/R (2)	26.0		278.0	7228
Case and Link Ejection Chute - UPR. L/R		6.0	277.0	1662
Ammunition Box - UPR. L/R		16.0	277.0	4432
Mountings and Misc. Equipment		49.9	270.0	13473
Ammunition 250 Rds.	177.0		273.0	48321
	407.8	75.9		
Total Useful Load Change	+483.7 lb.		157.6	+ 76260
Gross Weight Change	+524.2 lb.		166.6	87347
*GROSS WEIGHT, GUNFIGHTER	28003		454.8	12736133
GROSS WEIGHT CHANGES	524.2		166.6	87347
REVISED GROSS WEIGHT	28527.2		449.5	12823480

The gross weight c.g. is moved 5.3 inches or 3.7 percent MAC.

*Weight and Balance Report Number 24569 dated 1 June 1971.

FIGURE 2. Sample weight and balance breakdown for ECP's.

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3.6.4 Amended preproduction and production contract. If a contract is amended to provide for a major modification of one aircraft or group of aircraft, the contractor shall submit the following additional reports as soon as data are available (data previously submitted for the proposed change may be referenced and need not be duplicated):

a. Actual Weight Report - Intermediate Aircraft for the first modified aircraft, except that partial or complete detail weight statement and balance calculations as necessary shall be included to show differences over the detailed data furnished for the basic aircraft.

b. Revised sample Charts A and E, when applicable. If the modification to the aircraft does not require these charts to be changed, a written statement to that effect will fulfill this requirement.

c. The procuring activity will specify what additional actual weight reports are required when the contract is amended.

3.6.5 Contract for modification of new classes 1A, 1B, and 2 aircraft subsequent to acceptance and prior to service. The following requirements shall apply:

a. Sample Charts A and E shall be submitted if revision of the balance computer is necessary because of modifications incorporated.

b. Appendices shall be submitted to the weight reports for applicable blocks furnished under the production contract requirements. These appendices shall detail the modifications incorporated, their effect on weight empty, useful loads, gross weights, the extreme c.g. conditions, and the basic weight data. Representative aircraft shall be weighed after modification, preferably the same aircraft which were weighed under the production contract requirements, and an approved weighing form and derivation of the weight empty included in the appendices. The appendices shall be submitted not later than 30 days after the aircraft are weighed. Subject to approval by the procuring activity, the reweighing of aircraft may be omitted when changes in weight and balance are minor.

c. For classes 1B and 2 aircraft, the Weight Handbook shall be brought up to date. If charts are extensively revised, new charts shall be prepared and inserted.

d. Printed copies of released, or revised, Charts A and E shall be submitted as specified in 3.7.9.6.

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3.6.6 Contract for modification classes 1A, 1B, and 2 aircraft returned from service. The following requirements shall apply:

a. Sample Charts A and E shall be submitted if revision of the balance computer is necessary because of modifications incorporated.

b. Each aircraft shall be weighed after modification. Each aircraft shall be weighed prior to modification unless specifically waived by the procuring activity.

c. One copy of the weighing form, and the completed Charts A, C, and E for the 1st, 10th, 20th, and each subsequent 20th modified aircraft shall be submitted to the procuring activity not later than 30 days after the aircraft is weighed, with a listing of serial numbers, basic weights, arms, and moments (corrected to a standard typical inventory to permit comparison of weight and c.g. variations) for all aircraft modified during that period.

d. For classes 1B and 2 aircraft, the Charts A, C, and E data in the Weight Handbook shall be brought up to date. If the charts are extensively revised, new charts shall be inserted. A new handbook shall be prepared if the original has been mutilated or lost.

e. Subject to approval by the procuring activity, the requirements of 3.6.5 may be substituted for aircraft returned for modification after limited service.

3.7 Preparation of data. Each weight and status report shall be bound in a single folder with the data given in the order specified herein. The configuration of the aircraft for which data are submitted shall represent the official or contractual configuration since the original specification, as modified by changes of applicable effectivity. Pending changes shall not be included unless they have appropriate effectivity.

3.7.1 Weight and Balance Status Report. Unless otherwise specified, this report shall be submitted every month beginning immediately after the contract is let, and continuing on preproduction contracts until the Actual Weight Report is submitted for the aircraft as delivered to the procuring activity for final tests (for Navy, Part II and III demonstrations and the trials) and on production contracts until the Actual Weight Report for the first production aircraft is submitted. Additional submittals will be required by the procuring activity if weight or c.g. are considered to show an undesirable trend. The Weight and Balance Status Report (MIL-STD-1374, Part III) shall represent the status as determined by the contractor's running weight control records at the end of the period noted by the date on the form, and shall be submitted not later than 15 days after completion

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of the design phase represented by the report. Formal acceptance or approval by the procuring activity is required for the status reports. The following general instructions shall apply in their preparation:

a. In the initial issue of the status report, the contractor may reallocate weights among the weight groups of the original specification or the original guarantee weight empty, as deemed necessary from the best information available at the time, provided that specified weights of GFAE and the total specification or contract/guaranteed weight empty and useful load are unchanged. The purpose of this reallocation is to afford an opportunity for revising the contract/guaranteed weight breakdown to represent the then best available information on weight distribution and c.g. This distribution, as modified by later decreases and increases for Government responsibility changes and GFAE variations, will serve as the basis for determining under or overweight of the various groups. The group under or overweights are for comparative purposes only, since the contractor guarantees the total weight empty rather than the group breakdown. When such reallocations are made, complete information on all weight transfers among the groups, and their effect on c.g., shall be furnished as an appendix to the initial issue of the report. If the Estimated Weight Report has already been furnished, any major differences which will affect the use of the Estimated Weight Report as a reference source for arm and c.g. data shall also be included.

b. The initial issue of the status report shall include reference to the source of the recommended c.g. limits, brief indication of the critical flying qualities or structural considerations which dictated these limits, and reasons for any tapered or special c.g. restrictions. Similar data shall be furnished whenever limits are revised.

c. When mockup changes are introduced into the status report, complete information shall be submitted as an appendix to that issue, including brief descriptive identification of the changes, weights, arms, and moments, and a summary of their effect on the contract/guaranteed and predicted weight and balance of the aircraft.

d. A brief discussion shall be included with each issue of the status report describing the reasons for all major contractor responsibility weight variations and c.g. shifts, including any remedial measures being investigated or applied.

e. At the times when the current weight status and c.g. are brought into agreement with the Calculated Weight Report and the results of the actual weighing of the completed aircraft, appropriate note to this affect shall be made, and the Calculated or Actual Weight Report shall be referenced by number, date, and title.

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f. During the flight test phase, if changes in the current weight status in a status report period are less than 0.25 percent of the weight empty or 25 pounds, whichever is the larger, and the weight empty c.g. shift is less than 0.5 percent MAC for aircraft or 0.5 inch for rotorcraft, the status report for that period may be omitted and the procuring activity notified by letter; however, a report shall be submitted when the cumulative effect of changes since the last report reaches these values.

g. If a major modification is to be incorporated in one or a group of aircraft on the contract, information on the weight and c.g. status of the modified aircraft shall be included as an appendix to the status report. Status information furnished may be in the form of differences over the basic model, and shall continue as a separate report, if necessary, until the modified aircraft is delivered to the procuring activity for final tests (for Navy, Part II and III demonstrations and the trials).

3.7.2 Estimated Weight Report. This report shall be submitted after the contract has been awarded, but not later than 30 days after contract signing. Weights shown in this report shall be accurate and complete, and shall be in agreement with the original contract/guaranteed weight empty. The following data shall be included in the report in the order shown:

a. A table summarizing the weight, moments of inertia and c.g. (horizontal, lateral and vertical) data for the weight empty and all gross weight conditions covered in the balance calculations. Dimensional data (suitably illustrated by a small profile sketch of the aircraft) locating the horizontal and vertical moment axes and MAC or main rotor centerline shall be given below the table. The recommended c.g. limits shall be shown on, or tabulated below, the profile sketch. Refer to paragraph 3.7.1(b).

b. Group Weight Statement: (MIL-STD-1374, Part I)

c. Detail Weight Statement: (MIL-STD-1374, Part II) This statement is intended primarily to show that all power plant and equipment items have been considered; all groups, however, shall be detailed as completely as the data will permit at this stage of the design.

d. The following weight and balance calculations (Refer to Section 3.8):

(1) Weight empty detail balance calculations (including alternate alighting gear).

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(2) Gross weight balance calculations for all loading conditions listed in the aircraft contract detail specification, and for the most forward and the most aft loading conditions attainable as defined in this specification. When an aircraft can be operated with more than one type of fuel, data shall be included for all types.

(3) Derivation of the moment change owing to retracting the alighting gear.

(4) Weight, arm, and moment data for all alternate or special load items for which provision is made, and the effect on the primary or other applicable load condition.

e. Center of Gravity Envelope diagram showing a plot of weight vs c.g. similar to Figure 3 for all loading conditions listed in the specifications and for the most forward and aft loading condition attainabl to illustrate the effect of expendable and variable items of useful load and alighting gear retraction.

f. List of GFAE and weights.

g. Structural diagrams of wing, rotors (for rotorcraft), tail, body, nacelles, and alighting gear showing and identifying the following, as appropriate:

(1) Dimension from the reference datum for the horizontal and vertical moment arms to a convenient fixed point on the component.

(2) Scales in inches for determining horizontal and vertical dimensions.

(3) Principle structural stations and dimension to structural station 0 of the component.

(4) Location of the leading edge and length of the MAC or rotor centerlines.

(5) Surfaces, such as ailerons, flaps and other high lift devices, rudder, elevator, etc., with hinge lines shown in correct locations.

(6) Spar locations, major joints, and fold axes.

(7) Fuel tanks, wheel wells, bomb bays, and other major cutouts.

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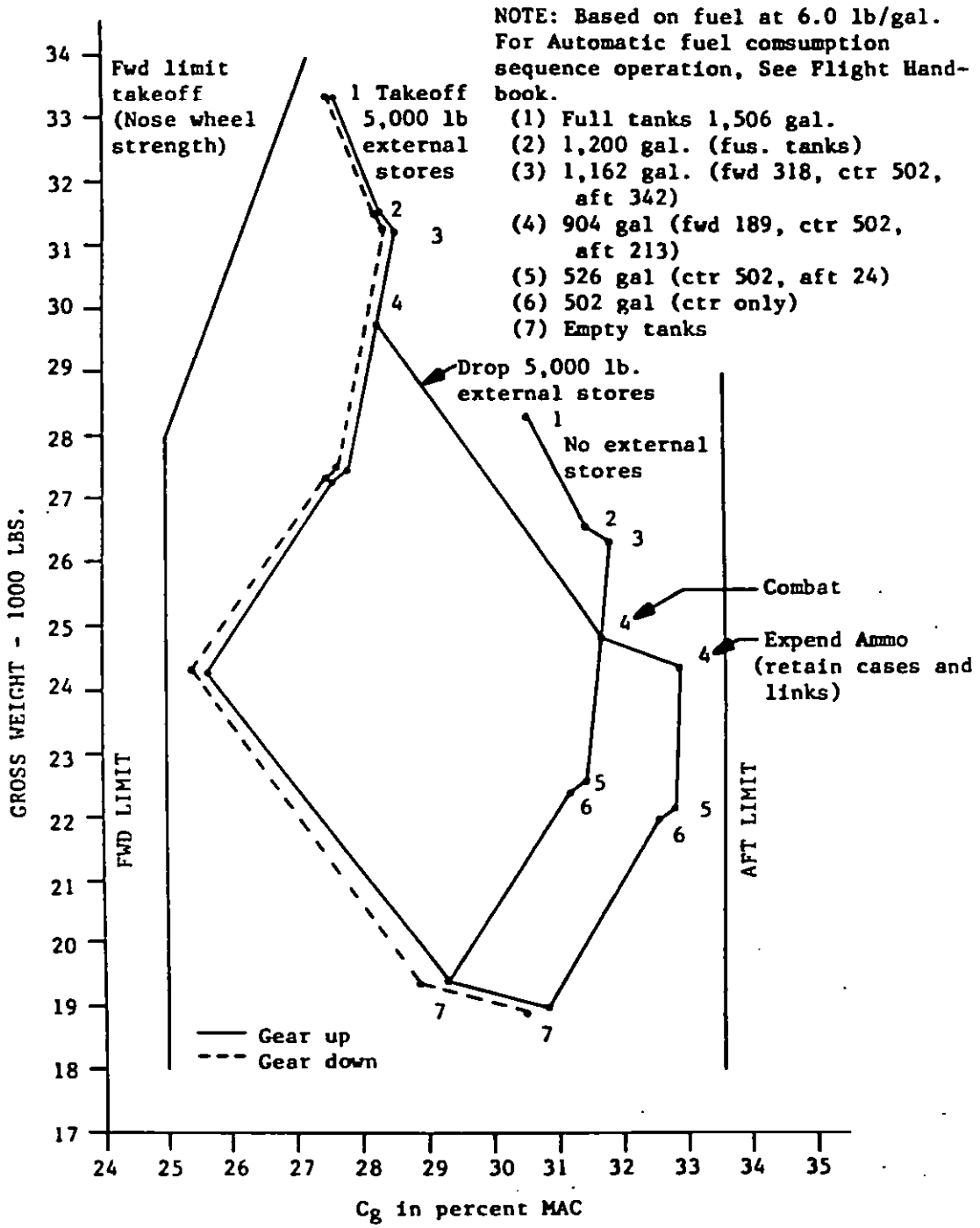


FIGURE 3. Sample center of gravity envelope diagram.

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(8) Centerlines of engine mounting points.

(9) Lengths, depths, and widths for body, nacelles, booms, and floats to identify the extent of the areas listed in the Group Weight Statement.

(10) Crosshatching of the outlines of wing and tail gross areas blanketed by the body, to identify the extent of the areas listed in the Group Weight Statement.

(11) For rotorcraft, an outline sketch of the transmission system, showing shafting and gearboxes from the engine through the system to the rotors, with design horsepower, torques, and gear ratios noted thereon for both 1/2 hour and continuous ratings.

3.7.3 Calculated Weight Report. This report shall be submitted for preproduction contracts approximately midway between the stages of design represented by the Estimated Weight Report and the first Actual Weight Report and for production contracts as soon as practicable after the contract has been awarded, but not later than 30 days after the design phase represented by the report. When Weight and Balance Status Reports are required, the Calculated Weight Report shall be coordinated with the current weight status and c.g. in the current status report, and the applicable status report shall be referenced by number, date, and title. Weights shown in this report shall be as accurate and as complete as possible with the information available. The following data shall be included in the report in the order shown:

a. A table summarizing the weight, moments of inertia and c.g. (horizontal, lateral and vertical) data for the weight empty and all gross weight conditions covered in the balance calculations. Dimensional data (suitably illustrated by a small profile sketch of the aircraft) locating the horizontal and vertical moment axes, and MAC or main rotor centerline, and reference to the applicable Weight and Balance Status Report shall be given below the table. The recommended c.g. limits shall be shown on, or tabulated below, the profile sketch. Refer to paragraph 3.7.1(b).

b. Group Weight Statement: (MIL-STD-1374, Part I).

c. Detail Weight Statement: (MIL-STD-1374, Part II).

d. The following weight and balance calculations (Refer to Section 3.8):

(1) Weight empty detail balance calculations (including alternate alighting gear).

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(2) Gross weight balance calculations for all loading conditions listed in the aircraft contract detail specification, and for the most forward and the most aft loading conditions attainable. When an aircraft can be operated with more than one type of fuel, data shall be included for all types.

(3) Derivation of the moment change owing to retracting the alighting gear.

(4) Weight, arm, and moment data for all alternate or special load items for which provision is made, and the affect on the primary or other applicable load condition.

e. Center of Gravity Envelope diagram showing a plot of weight vs c.g. similar to Figure 3 for all loading conditions listed in the specifications and for the most forward and aft loading conditions attainable, to illustrate the effect of expendable and variable items of useful load and alighting gear retraction.

f. A tabulation of specified and current weights of all GFAE included in the weight empty and determination of the net under or overweight of this equipment, and a separate tabulation of similar data for GFAE included in the useful load. All changes in the list as given in the original contract specification for the aircraft shall be explained by suitable comment and reference to change orders, ECP's, etc.

g. A tabulation of all authorized changes included in the reported weight empty and useful load, the weights and the groups to which the weights are allocated. Pending changes which have been incorporated shall also be listed separately. Changes shall be identified by the contract change letter, or change order number, designations, and by subject. Correspondence references or ECP numbers may be used for pending changes.

h. A tabulation deriving the contractor responsibility weight empty under or overweight, as follows:

(1) The original contract/guaranteed weight empty.

(2) The weight empty increases or decreases owing to the weights of (a) authorized changes, (b) pending changes, and (c) the net under or overweight of GFAE.

(3) The original contract/guaranteed weight empty as modified by (2) above.

(4) Comparison of the calculated weight empty with the revised contract/guaranteed weight empty derived in (3), to obtain the contractor responsibility under or overweight.

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i. Structural diagrams, corrected to date.

j. For production aircraft, a summary of changes and weight and balance differences over the prototype aircraft. When Weight and Balance Status Reports are not required, include information on any anticipated change in recommended c.g. limits, with reference to the source of the recommended limits, brief indication of the critical flying qualities or structural considerations which dictated these limits, and reasons for any tapered or special c.g. restrictions.

3.7.4 Actual Weight Report - First Aircraft. This report shall be submitted not later than 45 days after the aircraft is weighed, except that if the aircraft is assigned for test purposes, the report shall in no case be submitted later than the time of delivery to the testing activity. The following data shall be included in the report in the order shown:

a. Serial numbers of the aircraft for which this report is considered to be representative.

b. A table summarizing the weight, moments of inertia and c.g. (horizontal, lateral and vertical) data for the weight empty and all gross weight conditions covered in the balance calculations. Dimensional data (suitably illustrated by a small profile sketch of the aircraft) locating the horizontal and vertical moment axes, and MAC or main rotor centerline, shall be given below the table.

c. Group Weight Statement: (MIL-STD-1374, Part I).

d. Detail Weight Statement: (MIL-STD-1374, Part II).

e. Weighing Form: An approved weighing form, and when specifically requested by the procuring activity, Form AN-9250.

f. A detailed list of the items, weights, arms, and moments to be added to or subtracted from the as weighed condition to arrive at the weight empty data. A separate list of items, weights, arms, and moments shall be made for any test equipment or ballast installed in the aircraft at the time of weighing. For classes 1B and 2 aircraft, a separate list of items, weights, arms and moments added to the weight empty data to arrive at the basic weight moment shown in the Weight Handbook. Footnotes, including applicable letter references, shall call attention to any weight empty items authorized to be omitted at delivery, or temporarily omitted because of shortages or other reasons.

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g. All weight and balance calculations (refer to Section 3.8) from the Calculated Weight Report, corrected to agree with actual weight and c.g. data. Calculations for the alternate conditions not covered by the weighing shall be based on the results of actual weighing of the alternate alighting gear components, if available, or the alternate alighting gear data shall be identified as estimated or calculated. For seaplanes, the weights, arms, and moments for main and auxiliary beaching gear, as installed during a weighing, shall also be listed. Label data "calculated" if actual weights are not available. All calculations shall be given in at least as much detail as in the Calculated Weight Report. Actual weights shall be used for all contractor-furnished items and for the Government-furnished items installed by the contractor. Nominal (specified) weights shall be used for all other Government-furnished items.

h. Center of Gravity Envelope diagram showing a plot of weight vs c.g., similar to Figure 3 for all loading conditions listed in the specifications and for the most forward and aft loading condition attainable, to illustrate the effect of expendable and variable items of useful load and alighting gear retraction.

i. A tabulation of specified and actual weights of all GFAE included in the weight empty based on weighings of the items of equipment prior to installation in the first aircraft, and determination of the net under or overweight of this equipment. A separate tabulation of similar data for GFAE included in the useful load. All changes in the list as given in the original contract specification for the aircraft shall be explained by suitable comment and reference to change orders, ECP's, etc.

j. A tabulation of all authorized changes included in the reported weight empty and useful load, the weights, and the groups to which the weights are allocated. Pending changes which have been incorporated shall also be listed separately. Changes shall be identified by the contract change letter, or change order number, designations, and by subject. Correspondence references or ECP numbers may be used for pending changes.

k. A tabulation deriving the contractor responsibility weight empty under or overweight, as follows:

(1) The original contract/guaranteed weight empty.

(2) The weight empty increases or decreases owing to the weights of (a) authorized changes, (b) pending changes, and (c) the net under or overweight of GFAE.

(3) The original contract/guaranteed weight empty as modified by (2) above.

(4) Comparison of the actual weight empty with the revised contract/guaranteed weight empty derived in (3), to obtain the contractor responsibility under or overweight.

l. Unusable fuel and oil data as specified herein.

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m. Structural diagrams, corrected to date.

n. The Weight Handbook, including weighing form, Charts A, C, and E, (and Form DD-365F for class 2 aircraft). Chart A shall be checked to show the inventory applicable to the basic weight data derived in the report and entered in Chart C.

o. When the Weight Handbook is not required, information on:

(1) The type and location of leveling device and the means provided for taking measurements during weighings, in descriptive or diagrammatic form.

(2) The c.g. limits recommended by the contractor. For rotorcraft, if necessary, calculations shall be included converting limits in terms of inches from centerline of rotor to limits in terms of inches from reference datum.

3.7.5 Actual Weight Report - Intermediate Aircraft. This report shall be submitted not later than 45 days after the aircraft is weighed, except that if the aircraft is assigned for test purposes, the report shall in no case be submitted later than the time of delivery to the testing activity. The following data shall be included in the report in the order shown:

a. Serial numbers of the aircraft for which this report is considered to be representative.

b. A table summarizing the weight and horizontal c.g. data for the weight empty and the primary gross weight condition. (When changes are incorporated which materially affect vertical c.g. data submitted in a previous report, the weights, arms, and moments for the changes and their effect on the vertical c.g. shall be included in the balance calculations and in the summary.) Include lateral calculation when required.

c. Group Weight Statement: (MIL-STD-1374, Part I).

d. Detail Weight Statement (MIL-STD-1374, Part II) when required by the applicable Data Item Description (DID).

e. Weighing Form: An approved weighing form, and when specifically requested by the procuring activity, Form AN-9250.

f. A detailed list of the items, weights, arms, and moments to be added to or subtracted from the as-weighed condition to arrive at the weight empty data when required. For classes 1B and 2 aircraft a separate list of items, weights, arms, and moments added to weight empty data to arrive at the basic weight and moment shown in the Weight Handbook, when required. Footnotes, including applicable letter references, shall call attention to any weight empty items authorized to be omitted at delivery or temporarily omitted because of shortages or other reasons.

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g. The following weight and horizontal balance calculations (Refer to Section 3.8):

(1) Primary gross weight condition, based on the useful load in the last previous Actual Weight Report, corrected for changes incorporated.

(2) The alternate conditions not covered by the weighing. Calculations shall be based on the actual weighing of the aircraft and the latest information for the alternate alighting gear.

(3) Any additions or revisions to alternate or special load items, and effect on the primary or other applicable load condition, because of changes incorporated since the last previous Actual Weight Report.

h. A tabulation of all changes from the list of actual and specified weights of all items of GFAE included in the weight empty, as given in the last previous Actual Weight Report, and determination of the net under or overweight of this equipment, and a separate tabulation of similar data for changes in the list of GFAE in the useful load. Changes shall be explained by suitable comment and reference to change orders, ECP's, etc.

i. A tabulation of all authorized changes included in the weight empty and useful load subsequent to the last previous Actual Weight Report, the weights and the groups to which the weights are allocated. Pending changes which have been incorporated shall also be listed separately. Changes shall be identified by the contract change letter, or change order number, designations, and by subject. Correspondence references or ECP numbers may be used for pending changes.

j. A tabulation deriving the contractor responsibility weight empty, under or overweight, as follows:

(1) The original contract/guaranteed weight empty.

(2) The weight empty increases or decreases owing to the weights of (a) authorized changes, (b) pending changes, and (c) the net under or overweight of GFAE.

(3) The original contract/guaranteed weight empty as modified by (2) above.

(4) Comparison of the actual weight empty with the revised contract/guaranteed weight empty derived in (3), to obtain the contractor responsibility under or overweight.

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k. A tabulation of all actual changes in weight empty and useful load over the last previous Actual Weight Report, with weight distribution shown by groups, together with applicable references to authorized changes, contractor design changes, etc. The "manufacturing variation" shall also be listed. If the weight empty c.g. shift since the last previous Actual Weight Report exceeds 0.5 percent MAC for aircraft of 0.5 inch for rotorcraft, the moments caused by the changes and the unaccountable moment shall also be listed. If the cumulative weight empty c.g. shift since the first aircraft report, or since the last report in which the moments were listed, exceeds 1 percent MAC for aircraft or 1.0 inch for rotorcraft, similar moment data shall be furnished to account for the cumulative shift.

l. Unusable fuel and oil data as specified herein if a change in the fuel or oil systems has been made which materially affects the data previously determined.

m. Structural diagrams if the diagrams furnished in a previous report are modified because of changes incorporated.

n. The Weight Handbook, including weighing form, Charts A, C, and E (and Form DD-365F for class 2 aircraft). Chart A shall be checked to show the inventory applicable to the basic weight data derived in the report and entered in Chart C.

3.7.6 Appendices to Actual Weight Reports required prior to conducting procuring activity tests. These appendices are required only for Actual Weight Reports representative of one or more preproduction aircraft, or production blocks, from which any aircraft are allocated for procuring activity tests (for Navy, Parts II and III demonstrations and the trials). The appendices shall be submitted no later than the time of delivery of the aircraft to the testing activity.

3.7.6.1 Appendix to Actual Weight Report representative of the block. The following data shall be included:

a. A detailed tabulation of all modifications incorporated by the contractor subsequent to the basic report and prior to the delivery of the aircraft for tests, including effectivity points if not applicable to all aircraft of the block.

b. The effect of the modifications on actual and contract/ guaranteed weight and balance, contractor responsibility under or overweight, useful load, and extreme c.g. conditions. When pending changes have been included in under or overweight determination in the basic report, data shall be furnished to correct the tabulations to agree with the pending changes as finally authorized, including any necessary revisions to the under or overweight of GFAE.

c. Because of the above changes, revised pages of (or revisions to) the Weight Handbook data are required for all classes of aircraft.

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3.7.6.2 Appendix for instrumented aircraft. This appendix is required only for instrumented aircraft (for Navy applications, only for those instrumented aircraft allocated for structural, aerodynamic, and carrier suitability phases of the Parts II and III demonstrations and the trials). This appendix is considered as "Formal Data". Acceptance or approval by the procuring activity is required for this appendix . It shall include sufficient information to enable the testing activity to determine the actual as-delivered weight and balance of the aircraft and to assist in simulating loadings for test purposes. The following information shall be included:

a. Serial number of the aircraft, and test purpose(s) to which assigned.

b. Reference to the Actual Weight Reports for the block(s) from which the test aircraft were selected and which can be used as a reference source for representative weight and balance data for the "specification configuration."

c. Weighing Form: An approved weighing form, and when specifically requested by the procuring activity, Form AN-9250.

d. List of weight and balance corrections, if necessary, to reflect any changes between the actual weighing and the actual as-delivered condition.

e. Tabulations furnishing weight, arm, and moment data on the following, sufficiently detailed to enable test activity personnel to identify all components on board the aircraft as delivered:

(1) Test equipment.

(2) Ballast, including information indicating its function (simulation of specific missing equipment, over-all simulation of gross weights and c.g.'s, etc).

(3) Supplementary data as may be considered necessary by the contractor to show removal or addition of ballast or equipment to achieve the extremes of gross weight and forward and aft c.g., or loading combinations expected to be simulated on the instrumented aircraft during procuring activity tests (assuming, also, sufficient fuel for conducting the tests).

f. List of "specification configuration" equipment missing from the instrumented aircraft.

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3.7.7 Final appendices to Actual Weight Reports. These appendices are required only for Actual Weight Reports representative of blocks from which any aircraft have been allocated for procuring activity tests (for Navy, Parts II and III demonstrations and trials), and shall be submitted not later than 45 days after completion of tests. The data required for appendices in 3.7.6.1 shall be brought up to date to reflect all modifications incorporated and flight test changes found necessary for acceptance of the aircraft up to the time of completion of the tests.

3.7.8 Actual Weight Report - Last Aircraft. This report shall be submitted not later than 45 days after the aircraft is weighed, except that if the aircraft is assigned for test purposes the report shall in no case be submitted later than the time of delivery to the testing activity. The following data shall be included in the report in the order shown:

a. Serial numbers of the aircraft for which this report is considered to be representative.

b. A table summarizing the weight, moments of inertia and c.g. (horizontal, lateral and vertical) data for the weight empty and all gross weight conditions covered in the balance calculations. Dimensional data (suitably illustrated by a small profile sketch of the airplane) locating the horizontal and vertical moment axes, and MAC or main rotor centerline shall be given below the table.

c. Group Weight Statement: (MIL-STD-1374, Part I).

d. Detailed Weight Statement: (MIL-STD-1374, Part II).

e. Weighing Form: An approved weighing form, and when specifically requested by the procuring activity, Form AN-9250.

f. A detailed list of items, weights, arms, and moments to be added to or subtracted from the as-weighed condition to arrive at the weight empty data when required. For classes 1B and 2 aircraft, a separate list of items, weights, arms, and moments added to the weight empty data to arrive at the basic weight and moment shown in the Weight Handbook, when required. Footnotes, including applicable letter references, shall call attention to any weight empty items authorized to be omitted at delivery or temporarily omitted because of shortages or other reasons.

g. All weight and balance calculations (refer to Section 3.8) similar to those furnished in the Actual Weight Reprot - First Aircraft corrected to agree with the weight and c.g. of the last aircraft and including all changes in the useful load and alternate or special load items. Calculations for the alternate conditions not covered by the weighing shall be based on the results of actual weighing of the alternate alighting gear.

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h. Center of Gravity Envelope diagram showing a plot of weight vs c.g., similar to Figure 3 for all loading conditions listed in the specifications and for the most forward and aft loading condition attainable, to illustrate the effect of expendable and variable items of useful load and alighting gear retraction.

i. A tabulation of all changes from the list of actual and specified weights of all items of GFAE included in the weight empty, as given in the last previous Actual Weight Report, and determination of the net under or overweight of this equipment. A separate tabulation of similar data for changes in the list of GFAE included in the useful load. Changes shall be explained by suitable comment and reference to change order, ECP's, etc.

j. A tabulation of all authorized changes included in the weight empty and useful load subsequent to the last previous Actual Weight Report, the weights and the groups to which the weights are allocated. Pending changes which have been incorproated shall also be listed separately. Changes shall be identified by the contract change letter, or change order number, designations, and by subject. Correspondence references or ECP numbers may be used for pending changes.

k. A tabulation showing the contractor responsibility weight empty under or overweight, as follows:

(1) The original contract/guaranteed weight empty.

(2) The weight empty increases or decreases owing to the weights of (a) authorized changes, (b) pending changes, and (c) the net under or overweight of GFAE.

(3) The original contract/guaranteed weight empty as modified by (2) above.

(4) Comparison of the actual weight empty with the revised contract/guaranteed weight empty derived in (3), to obtain the contractor responsibility under or overweight.

l. A tabulation of all actual changes in weight empty and useful load over the last previous Actual Weight Report, with weight distribution shown by groups, together with applicable references to authorized changes, contractor design changes, etc. The "manufacturing variation" shall also be listed. If the weight empty c.g. shift since the last previous Actual Weight Report exceeds 0.5 percent MAC for aircraft of 0.5 inch for rotorcraft, the moments caused by the changes and the unaccountable moment shall also be listed. If the cumulative weight empty c.g. shift since the first aircraft report, or since the last report in which the moments were listed, exceeds 1 percent MAC for aircraft or 1.0 inch for rotorcraft, similar moment data shall be furnished to account for the cumulative shift.

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m. Unusable fuel and oil data as specified herein if a change in the fuel and oil systems has been made which materially affects the data previously determined.

n. Complete set of updated structural diagrams.

o. The Weight Handbook, including weighing form, Charts A, C, and E (and Form DD-365F for class 2 aircraft). Chart A shall be checked to show the inventory applicable to the basic weight data derived in the report and entered in Chart C.

3.7.9 Handbook, Weight and Balance Data, (classes 1B and 2 aircraft). The contractor shall prepare and reproduce charts and complete the Weight Handbook for each aircraft in accordance with the instructions given herein. (For Air Force contracts further information may be found in T.O. 1-1B-50 and MIL-M-5920.)

3.7.9.1 General. The following general instructions apply:

a. Insert all aircraft identifying data on the title page and the various charts and forms.

b. All moment data entered in the handbook shall be divided by the constant 1,000. A constant of 100 for extremely small aircraft, or 10,000 for extremely large aircraft, may be used subject to prior approval by the procuring activity. Wherever "moment" is referred to in the following instructions to the contractor for preparation of chart data, it shall be understood to mean "moment/constant."

c. Each Weight Handbook shall represent the as-delivered condition of the aircraft for which it is prepared, and include a sample Form DD-365F for class 2 aircraft. Subject to specific approval of the procuring activity in each case and dependent upon the rate of production, handbooks for class 1B aircraft may be prepared for groups of identical aircraft. In the latter case, the basic weight and moment shall be based on actual data applicable to the group and on an assumed primary operating condition, with the "Delivery Equipment" column on Chart A dated above the column to agree with the Chart C entry, and checked to agree with the equipment in the assumed condition; a copy of Figure 4 shall be inserted in the front of the handbook of each aircraft of the group, and the contractor, or the activity responsible for loading for flyaway delivery, shall insure that all aircraft are properly loaded for such flyaway delivery.

d. The contractor shall assign and place the date of preparation (or revision) in an inconspicuous location on each page of the charts (e.g., in the binding margin, etc.) Charts A and E shall conform to a page size of 8-1/2 by 11 inches, including approximately 1-inch space allowance for binding, and the following in the upper outer corner of all pages of the respective charts:

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IMPORTANT NOTICE

Subj: Handbook of Weight and Balance Data for Model _____Aircraft, Basic Weight Data for.

1. The basic weight data entered in Chart C of the Weight Handbook of this aircraft applies to an assumed primary operating condition. The "Delivery Equipment" column in Chart A has been checked to show the equipment included in the basic weight for this assumed condition and, therefore, does not necessarily represent the equipment actually installed in this aircraft.

2. The weight and balance personnel of the activity receiving this aircraft shall correct the basic weight to agree with the actual operating condition of the aircraft in accordance with the following instructions:

(a) After installation of the required equipment conduct a Chart A inventory in the "Check 1" Column.

(b) Compare this inventory with that for the basic weight entry in Chart C. Correct the Chart C entry as necessary for any items of equipment which have been added or removed.

3. After completion of the above, this sheet shall be removed from the handbook.

FIGURE 4. Notice to be inserted in the Weight Handbook when an assumed primary operating basic weight condition is entered on Chart C.

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On Chart A:

Chart Date - See Page 1

On Chart E:

Page _____ of _____
Model _____
Chart Date - See Page 1

On page 1 of Charts A and E, the words "See Page 1" shall be replaced by the most recent date of the individual preparation or revision dates of the pages comprising that chart.

e. Chart E shall be inserted immediately after the Chart C pages.

f. The instruction book furnished with each balance computer shall be inserted at the end of Section III.

g. In no case shall an item be listed on both Charts A and E.

3.7.9.2 Chart A, Form DD-365A. Instructions for the preparation of Chart A are furnished below:

a. Operating equipment which is or may be installed and for which provision or fixed stowage has been made shall be listed on Chart A as separate items, or groups of inseparable items, suitably identified in order to facilitate an inventory of equipment by inexperienced personnel. If an item can be in a "stowed" and an "installed" location, the item shall be listed for both locations and so labeled. All items weighing 2 pounds or more shall be listed for aircraft whose weight empty is under 25,000 pounds, and items weighing 5 pounds or more, for larger aircraft. Figure 5 furnishes a list of various items to consider for Chart A.

b. Aircraft compartments shall be designated by capital letters and appropriate descriptive nomenclature. The compartment letter designation and name shall be shown at the top of each list of equipment items for one compartment. The compartment designation shall be underlined and separated from the equipment list by one blank line. The limits of each compartment in inches from the reference datum shall be placed on same line as the compartment designation. These compartment limits shall agree with those shown on Chart E. Equipment located external to the body compartments, e.g., in wings, nacelles, etc, shall be listed at the end of Chart A under appropriate designations. When required by the procuring agency, sketches of Chart A item locations will be inserted as a facing page to the corresponding items listing as shown in Figure 5 (Sheet 4 of 4).

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The following list, representative of types of items which should be tabulated on Chart A, includes equipments which normally have a fixed location and which might be in or out of the aircraft during weighing or for various operating configurations. Items should be listed on Chart A only if they weigh 2 pounds or more for aircraft under 25,000 pounds weight empty and weigh 5 pounds or more for larger aircraft. Items should be listed by descriptive name or type, part number or equipment designation, quantity, capacity, and other appropriate means to avoid ambiguity and to facilitate identification and inventory by personnel who may not be thoroughly familiar with the aircraft. Equipments which are alternates to each other shall be suitably identified, e.g., "(alternate to item A-21)" or similar.

A. Alternate or operating equipment:

1. Autopilot removable components.
2. Electrical power supply, e.g., batteries, generators, inverters, etc.
3. Air intake filters.
4. Water injection tanks.
5. Engines and propellers when alternate installations are provided or when weight variations encountered materially affect aircraft weight and center of gravity.
6. Hydraulic fluid over normal system capacity.
7. Hand cranks if not built in.
8. Pneumatic bottles for operating mechanisms.
9. Removable main and auxiliary fuel and oil tanks and tanks which are droppable only in emergencies. Tanks normally dropped during the mission should be listed on Chart E.
10. Fuel vapor inertion generators or cylinders.
11. Anchor gear.
12. Permanent ballast.
13. Catapult and arresting hooks and arresting hook point.
14. Portable bilge pump and hose.
15. Unusable fuel and unusable oil.
16. Auxiliary power unit.
17. Windshield wipers and washer or anti-icing fluid.
18. Alternate alighting gears.
19. Mooring gear, covers, etc.
20. Ladders and platforms.
21. Tool kits.

FIGURE 5 (Sheet 1 of 4). Typical items to be listed on Chart A.

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B. Crew safety and comfort equipment not integral with structure:

1. Heating, cooling, and pressurizing equipment and insulation, e.g., heaters, heat exchangers, blowers, etc.
2. Oxygen equipment, e.g., generators, cylinders, weight of contents, regulators, etc.
3. Parachutes if fixed stowage provided. When no fixed stowage is provided and parachute is worn by crew member in flight, include in crew weight on Chart E.
4. Life rafts and emergency equipment stowed with rafts. When individual rafts are attached to parachute pack and worn by crew member in flight, list on Chart E.
5. Crew seats, safety belts, bunks, beds, cushions, tables, mirrors, etc. For ejection seats, list ejectable portion of seat and state items included.
6. Lavatory and toilet facilities.
7. Curtains.
8. Anti-icing removable components, e.g., boots and attaching parts, heaters, tanks, fluid.
9. Flares, tubes, signal pistols and ammunition, and other pyrotechnic equipment. When the quantity of flares materially affects aircraft weight and center of gravity, list on Chart E.
10. Navigation equipment, e.g., driftsight, navigational kit or packet, sextant, astrocompass, etc.
11. Fire extinguisher cylinders.
12. Emergency axes, first aid kits, etc.
13. Blind flying hood.
14. Electronic equipment, e.g., receivers, transmitters, dynamotors, antennas, etc. Mounts shall be listed separately.

C. Protection:

1. Removable armor and deflector plate.
2. Removable bullet resistant glass.

D. Armament installations:

1. Guns, chargers, blast tubes, sights, and removable mounts.
2. Ammunition boxes, chutes, boosters.
3. Turrets.
4. Gun cameras.

FIGURE 5 (Sheet 2 of 4). Typical items to be listed on Chart A.

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5. Racks, shackles, adapters, pylons, etc., for stores, rockets, etc. When these items are a variable with the stores carried, list on Chart E with appropriate explanatory information.
6. Bomb sight.
7. Electronic equipment.

E. Personnel and cargo transport equipment:

1. Troop and passenger seats and belts.
2. Tables.
3. Window curtains, rods, rugs, removable soundproofing and insulation, and other interior furnishings.
4. Food and drink containers. List contents separately.
5. Stoves, hot plates, refrigerators.
6. Utensils and tableware (stowed location).
7. Cup containers.
8. Litters and supports. List stowed and installed locations.
9. Tie down gear (stowed location).
10. Special floors.
11. Ramps.
12. Hoists, winches, monorails, dollies.

F. Photographic equipment for photographic aircraft:

1. Removable components providing for camera installations, e.g., intervalometers, mounts normally carried at all times, etc. Camera, magazines, film, flashbombs, hatches, and mounts variable with the camera installation shall be listed on Chart E.

FIGURE 5 (Sheet 3 of 4). Typical items to be listed on Chart A.

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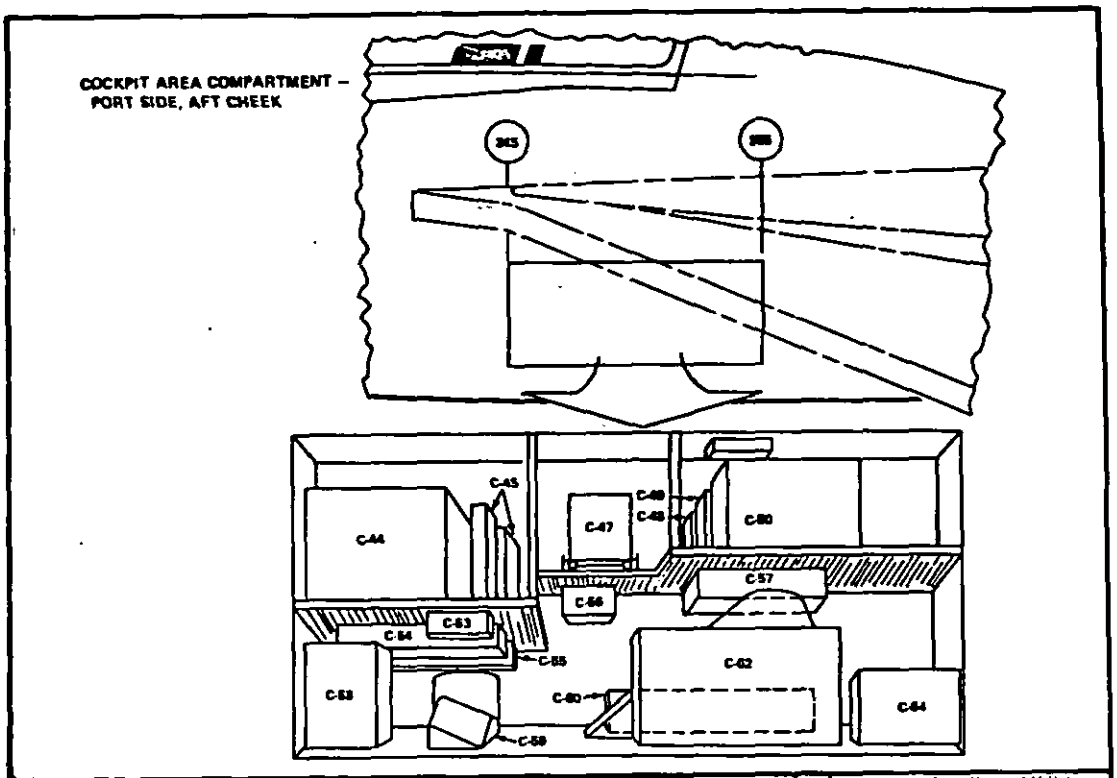


CHART A - BASIC WEIGHT CHECK LIST

PAGE 5 OF 14 PAGES | AIRPLANE MODEL F-14A | SERIAL NUMBER

COMPARTMENT AND ITEM NUMBER	ITEMS and LOCATION (Grouped by Compartment)		WEIGHT	ARM	MOMENT / 1000
C	COCKPIT AREA COMPARTMENT (ARM 313.8 – 365.0)				
	Aft Cheek – Port Side (Arm 345.0 – 365.0)				
C-44	Converter, Analog Data	CV-2784/AVA-12	53	353	18.7
C-45	Preamplifiers (2)	AM-6368 A/ALR-45	7	350	2.5
C-47	Assy, Amplifier/Relay	AM-3834/ARA-50	5	364	1.8
C-48	Synchronizer	SN-41BA/APX-76	7	379	2.6
C-49	Beacon, Amplifier	SA-1988A/APX-76	9	374	3.4
C-50	Recr, Aux UHF	R-1289/ARR-69	7	374	2.6
C-53	Pwr Supply, Fuel Flow	ECK-85-A3744	(Ref)	–	–
C-54	Set, Airborne Test	TS-1913A/APX	(Ref)	–	–
C-55	Switch, Latching	SA 1474/A	(Ref)	–	–
C-56	Manager, Ground Pwr	M 340271-2	(Ref)	–	–
C-51	Decoder, Pulse	KY-851/ARA-63	7	377	2.6
C-58	Recr, Transponder	RT-859/APX-72	15	381	5.7
C-59	Assy, Rate Gyro	TRU-3A/A	(Ref)	–	–
C-60	Recr, ECM Radar	R-1794A/ALR-66	13	379	4.9
C-63	Analyzer, Elec Pulse	TS-3053A/ALR-45	23	378	8.7
C-64	Blanker, Interference	MX-6957/A	6	382	2.3

DELIVERY DATE | RECORD OF CHECKING (Enter date)

CHECK 1 2 3 4 5 6 7 8 (WT. ARM/100, CHART C ENTRY)

CHART DATE: See page 1

1/ Enter constant used below line.

DD FORM 1 SEP 84 365A

FIGURE 5 (Sheet 4 of 4). Chart A facing page illustration.

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c. The order in which items are listed shall be, insofar as is practicable, such that the moment arms increase progressively from the forward limit to the aft limit of the compartment. If a floor or partition divides a compartment into distinct sections, the Chart A items for that compartment shall be listed by sections. No item or group of items shall be listed in a compartment unless the c.g. of the item or group falls within the limits of the compartment. Items shall be numbered consecutively by compartments, e.g., items in compartment A shall be designated A-1, A-2, A-3, etc.

d. Unusable and trapped fuel and oil (weights, arms, and moments) shall be listed as two separate items, and the drain points identified by suitable notes such as "when drained in normal ground attitude from (identity of drain point)." When "unusable" and "trapped" fuel and oil are different, this entry will be so noted and the difference shall be shown on the fuel and/or oil pages in the Chart E's and a note in the weighing instruction concerning this difference shall be placed therein.

e. For convertible aircraft with alternate alighting gear, data shall be listed for the alternate alighting gear conditions, if materially different, and "in normal beaching gear attitude" or similar expression used where appropriate.

f. Weights and moments shall be listed to the nearest whole number, and the arms shall be listed to one decimal place.

3.7.9.3 Weighing form, Form DD-365B. The contractual weighing schedule provides for weighing certain aircraft during the contract. The data from these weighings shall be utilized in preparing weighing forms for the Weight Handbook. For those aircraft not actually weighed, the weighing data for the weighed aircraft of the same production block shall be used and a note added in the "Remarks" block, "Based on the actual weighing of aircraft serial No. _____." Since this procedure presupposes identical aircraft with normal production tolerances on weight and c.g., any known differences shall be accounted for in determining the basic weight for aircraft not actually weighed.

3.7.9.4 Chart C, Form DD-365C. The as-delivered basic weight, and moment (alighting gear extended) and, where applicable, the balance computer index, shall be listed. The "assumed primary operating condition" basic weight data shall be entered when the deviation permitted by 3.7.9.1c is authorized. If the "assumed primary operating condition" includes the weight of any shortage items which are not listed normally on Chart A as inventory items, these items shall be listed as Chart C entries to avoid confusion when the items are subsequently installed in service. The date and data entered on Chart C shall agree with the date and inventory marks entered in the "Delivery Equipment" column on Chart A.

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3.7.9.5 Chart E. Chart E shall provide means for determining quickly and accurately the horizontal c.g. position of the aircraft under any loading condition. When loading limitations or cautions are necessary to avoid adverse lateral load distribution, pertinent information shall be included in the appropriate tables. The requirements specified herein shall be regarded as minimum, and the contractor shall include such additional data as may be deemed necessary to insure that Chart E will fulfill adequately its intended purposes. Data shall be presented in the form of tables, and in the order listed below insofar as practicable consistent with space arrangement and use of minimum number of pages. Weight and moment data shall be listed to the nearest whole number. Arms shall be listed to one decimal place. For sample data, see Figures 6 through 21, inserted where applicable (see Numerical Index).

3.7.9.5.1 Weighing instructions. These data shall present any special weighing instructions or information which are peculiar to the individual aircraft and which are necessary for field weighing personnel who are unfamiliar with these aircraft, to reproduce with reasonable accuracy the conditions necessary for determining the basic weight. (It is not intended that general weighing preparation and techniques be described.) The weighing instructions shall also include a required weighing frequency for the aircraft that has been determined by consultations between the contractor and the procuring activity. Other typical items are as follows:

a. Open or closed, retracted or extended position of sliding nose compartment cowl, canopy, slats, leading or trailing edge flaps, etc., if they would individually or cumulatively materially affect basic weight c.g. Give moment/constant correction if weighed with item in opposite position.

b. Cautions pertaining to draining of water from bilges of seaplanes or amphibians before weighing.

c. Brief defueling and oil drainage instructions, if it is necessary to prescribe a defueling procedure to accomplish a known condition of drainage in ground attitude (i.e., "trapped" fuel and oil aboard). Such procedures may involve a given sequence of defueling, transfer from certain tanks using boost pumps where single point defueling is provided, maintenance of certain suction pressures on defueling pumps, etc. In addition, where "trapped" and "unusable" fuel and oil and materially different quantities, the following notes shall be completed and included in the instructions:

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WEIGHING INSTRUCTIONS

1. Aircraft Condition

   A. The basic weight condition is established with wheels down, slats extended, and with the nose cone and canopy closed. If weighed with the nose cone open, the moment/1,000 correction is minus (-) 4.8. If weighed with the canopy open, the moment/1,000 correction is minus (-) 3.3.

2. Fuel Draining

   A. Operate boost pumps in forward fuselage and aft center section tanks. Check "no flow transfer" warning light in cockpit to insure that transfer system is empty.
   B. Defuel in three-point attitude through pressure fueling and defueling valve located on bottom centerline of the aircraft at station 272.5 using the external defueling pump. Do not open water drains in lines or fuel strainers. Caution: Aft fuselage tank drains by gravity into the main fuselage tank and thence to the defueling valve. Special care must be taken to insure that the aft fuselage tank is fully drained as follows:

      (1) If using the self-priming pump, allow pump to run for 5 minutes after flow initially stops to make sure that gravity flow from aft fuselage tank is completed.
      (2) If using the non-self-priming pump, limit pump suction to a pressure of 1.0 psi to keep pumping rate below gravity flow rate aft fuselage tank, thus avoiding loss of pump prime before aft fuselage tank is drained.

   C. Fuel remaining aboard after defueling in the above manner is the unusable fuel included in aircraft basic weight (see Chart A for quantity).

3. Leveling

   A. Plumb bob leveling suspension fitting and target are located in the nose gear bay on the upper and lower longerons on the LH side. See detail on page 2.
   B. To reduce the large aircraft ground angle for ease in leveling, blow up the main gear struts to fully extended position; deflate and tie the nose gear strut in fully compressed position.
   C. Center the nose wheels and make sure that they are free to roll on the floor when the wing jacks are extended. Place standard jacks under the wing jack pads and raise aircraft until approximately level. (Electric weighing cells must never be on the jacks for this operation.)
   D. Shore up main wheels with wooden cribbing. Lower wing jacks until aircraft again rests on wheels.
   E. Place electric weighing cells on jack at forward fuselage jack point and on wing jacks. Raise jack until wheels are clear, leveling aircraft with jacks.
   F. Alternate to item C and D above: Raise aircraft with overhead hoist until approximately level. Proceed with item E.
   G. When weighing on wheels with mechanical scales, raise aircraft by method C or F until slightly nose down in attitude; then, if necessary, use jack at forward fuselage jack point in order that the scale may be inserted under nose wheels. Shore up main wheels with cribbing on top of scales until aircraft is almost level with wheels resting on the cribbing; bleed main gear struts as required for the final level adjustment.

4. Measuring

   A. The "jig point" is the forward fuselage jack point, fuselage structural station 266.09. See detail on page 2. Caution: Do not confuse the jack point socket with the boresighting fixture socker located a few inches farther forward in the nose gear bay.
   B. For "letter" dimensions referred to below, see the diagram on page 2.
   C. When weighing on jack points, actual measurements during weighing are not necessary. After leveling, the following dimensions for "E" and "F" may be inserted directly in the weighing form:

      (Wing) Aft jack point to reference datum. Dimension E=422.54
      (Fuselage) Forward jack point to reference datum. Dimension F=266.09

   D. When weighing on wheels, measure dimensions "B" and "D" during weighing and after leveling. Using these actual dimensions and the jig point dimension "I", determine "E" and "F". For checking purposes, approximate dimensions "E" and "F" are given in the table on page 2.

FIGURE 6. Sample weighing instructions.

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"The Basic Weight is established with 'unusable' fuel and oil (equivalent to the zero point of the gages). If the aircraft is weighed after draining fuel and oil in normal ground attitude as outlined above, the following increments shall be (added) (subtracted) to the 'as-weighed' condition to obtain the Basic Weight:

Item	Weight	Moment/constant
Fuel Oil		

If the aircraft is weighed with completely dry fuel and oil systems, the 'unusable fuel' and 'unusable oil' weight and moment listed on Chart A shall be added to the 'as-weighed' condition to obtain the Basic Weight."

d. Suggested procedure for leveling if the normal ground attitude or alighting gear arrangement is such that placing the aircraft in level attitude for weighing might entail unusual problems for field personnel. In this category are aircraft which have extremely long nose gear struts and require extreme rotation of attitude, aircraft with bicycle gear which might require a sequence of steps of jacking, using several sets of jack points, etc. Include brief statement of recommended shock strut inflation or deflation, approximate height main or nose wheels must be raised, lift points available for hoisting to level position, any cautions considered necessary, etc.

e. Measuring data, if descriptive instructions are necessary for specific location and positive identification of jig points.

f. Sketches, tables, etc., when necessary, to amplify the weighing information required for the aircraft diagram.

g. For aircraft which may be weighed on removable beaching gear, list "dry weights" of beaching gear, moment arms, and moment/constant for use in correcting "as weighed" to Basic Weight. If dry weight varies sufficiently to affect the accuracy of the weighing, add a note: "Approximate. See actual weights stenciled on beaching gear."

3.7.9.5.2 Aircraft diagram. The aircraft diagram shall consist of a side elevation of the aircraft and a plan view of the body and one wing showing the data listed below. The wing diagram may be broken off outboard of the point required to show wing equipment and loading installations. Alternate alighting gears shall be represented by separate diagrams if necessary for clarity.

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a. Reference datum.

b. Body stations, with dimension from reference datum to structural Station 0, if different from the reference datum, and identifying principal structural stations.

c. Leveling lugs, or device: Identify type and specific location.

d. Main bulkheads, crew stations, the prominent items of equipment suitably labeled to provide easy identification. Equipment need be shown only in sufficient detail to enable field personnel to determine relative locations. For transport type aircraft, additional plan view diagrams shall be included to show alternate passenger or troop seat, or litter arrangement.

e. A scale in inches to facilitate the determination of approximate horizontal moment arms from the reference datum.

f. The length of the MAC and the distance from its leading edge to the reference datum. Incidence of the MAC shall not be shown. For rotorcraft the distance from the reference datum to the centerline of the main rotor shall be given.

g. Location, and distance in inches from reference datum, of suitable jigged points or fittings from which measurements may be taken to locate the reactions during weighings. Identify as "Dimension I." Jack pads may be used when they are jigged. These points shall be described to facilitate ready and accurate location in the field, or shown diagrammatically by an enlarged detail.

h. Landing or beaching gear dimension data: Show dimension lines and identify as follows:

(1) Jig point to centerline of main wheels, "Weighing Dimension B."

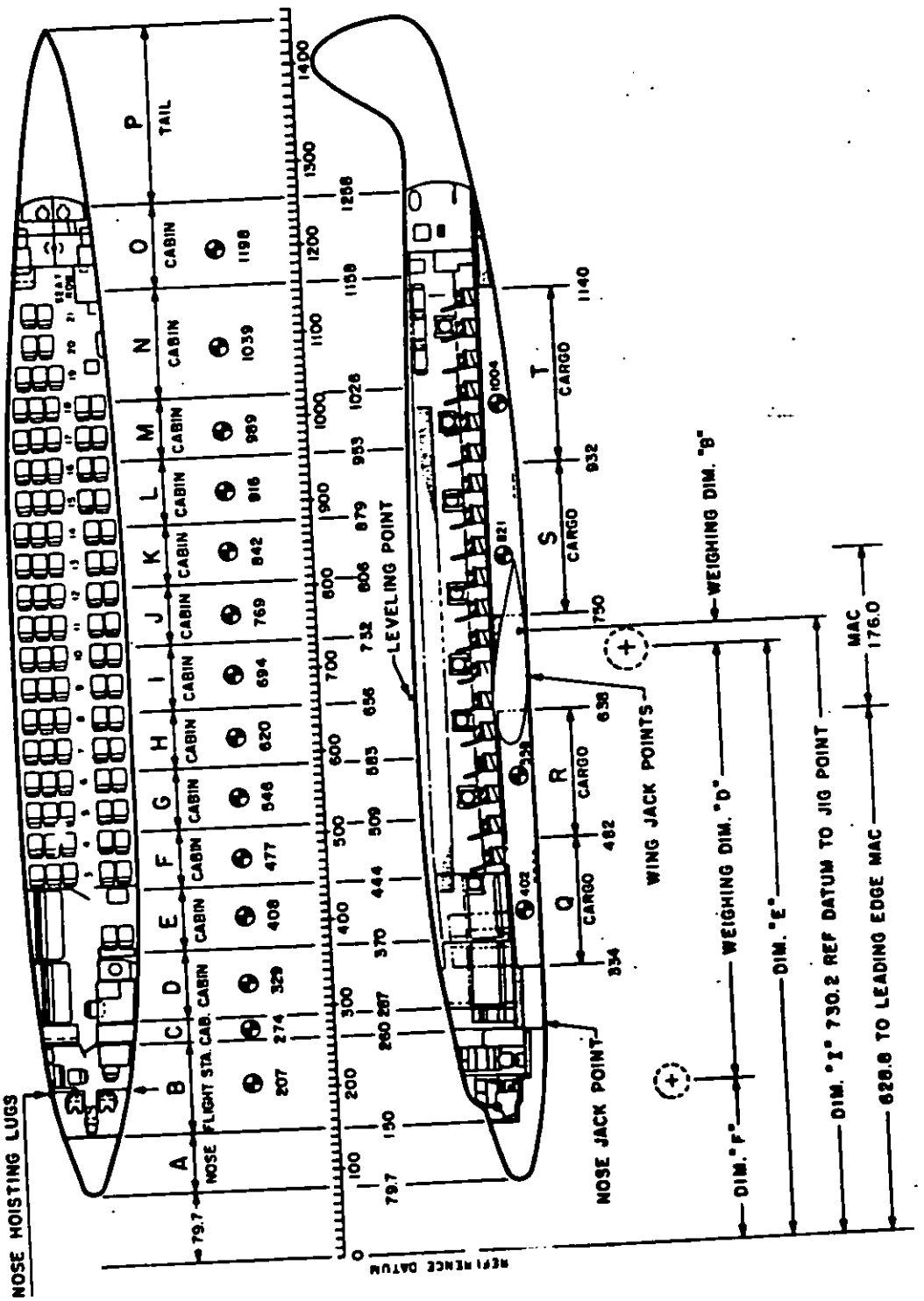
(2) Distance between centerlines of main and nose or tail wheels, "Weighing Dimension D."

(3) Reference datum to centerline of nose or tail wheel, "Dimension F."

(4) Reference datum to centerline main wheels, "Dimension E."

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AIRCRAFT DIAGRAM



NOTE: REFER TO WEIGHING INSTRUCTIONS ON PAGE 1 FOR DETAILS AND DIMENSIONS OF HOISTING LUGS, JIG POINT, JACK POINTS, AND LANDING GEAR WHEELS.

FIGURE 7. Sample aircraft diagram for class 2 aircraft.

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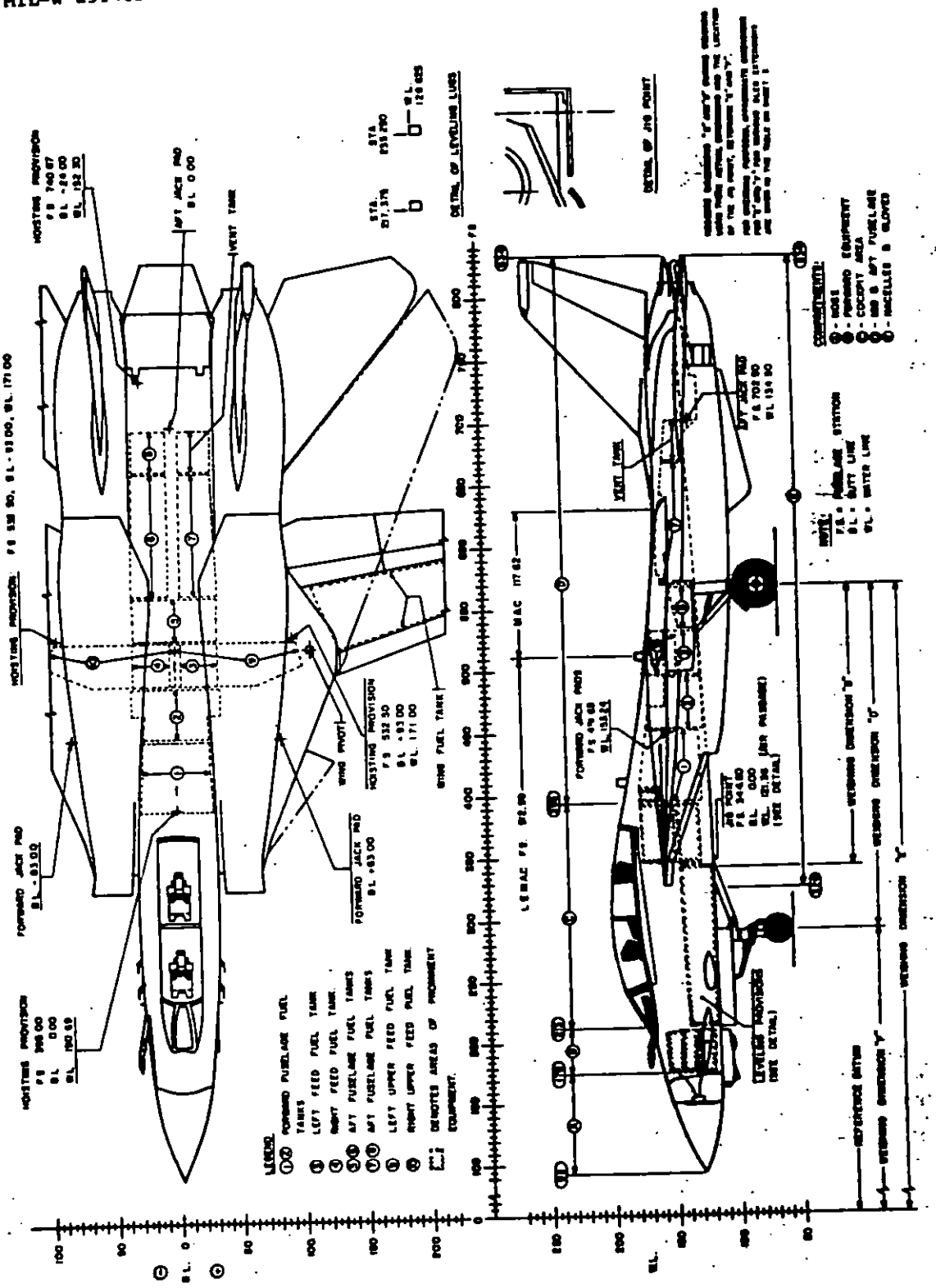
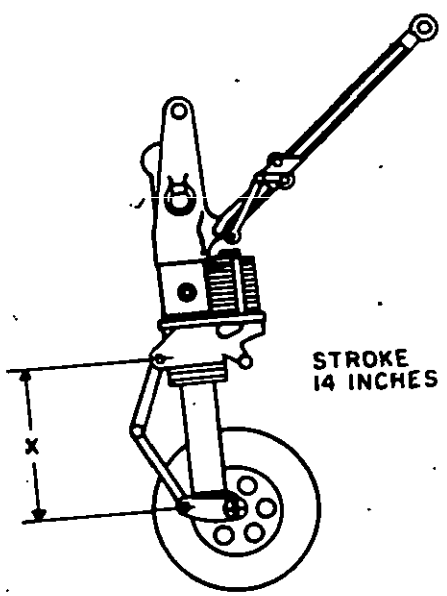


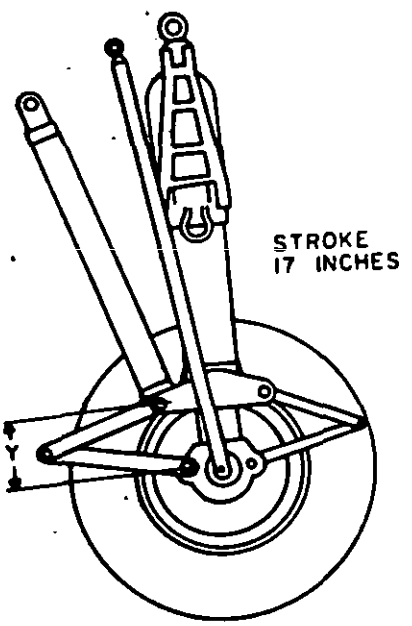
FIGURE 8. Sample aircraft diagram for class 1A and 1B aircraft.

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LANDING GEAR DIMENSIONS

NOSE WHEEL			MAIN WHEELS		
X		* DIM. F	Y		* DIM. E
Fully Compressed	5.32	69.08	Fully Compressed	4.81	229.79
	6.32	69.15		5.81	229.86
	7.32	69.23		6.81	229.93
	8.32	69.31	Static Condition	7.81	230.00
	9.32	69.38		8.81	230.07
	10.32	69.47		9.81	230.14
	11.32	69.55		10.81	230.21
	12.32	69.62		11.81	230.28
	13.32	69.70		12.81	230.35
	14.32	69.78		13.81	230.42
	15.32	69.86		14.81	230.48
	16.32	69.94		15.81	230.55
Static-Fully Extended	17.32	70.02		16.81	230.62
				17.81	230.69
				18.81	230.76
				19.81	230.83
				20.81	230.90
			Fully Extended	21.81	230.97





* APPROXIMATE. DETERMINE ACTUAL DIMENSIONS DURING WEIGHING.

RH GEAR LOOKING OUTB'D

FIGURE 9. Sample landing gear dimension data.

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(5) Add an asterisk to all four dimensions and the following asterisk note in a conspicuous place on the page: "*Measure dimensions 'B' and 'D' during weighing. Using these actual dimensions and the jig point dimension 'I', determine 'E' and 'F'. For checking purposes, approximate dimensions 'E' and 'F' for various oleo extensions are given in the table on page _____."

(6) Include a table on the referenced page listing dimensions 'E' and 'F' corresponding to various shock strut extensions from fully compressed to fully extended positions. Label the table: "Approximate. Determine actual dimensions during weighing."

(7) Include a detail of the shock strut to identify specifically where the shock strut extensions are to be measured. The contractor shall verify the accuracy of these dimensions during the weighings required by the contractual weighing schedule and advise the procuring activity of the shock strut extension and landing gear dimensions obtained.

i. Show all additional points, such as landing gear and wing jack points, lift tubes, tail hoist fittings, etc., which might be used as reaction points during weighings together with distances to the reference datum. Identify as "Dimension E" or "Dimension F," as appropriate. For these dimensions (except for jack points on landing gears), add a note: "Actual measurements during weighing are not necessary. After leveling the aircraft, these dimensions for 'E' and 'F' may be inserted directly in the weighing form."

j. For class 1A and 1B aircraft only, stations or compartments where miscellaneous equipment for ferry flights can be carried, including the maximum allowable weight and the moment for each 10 pounds of weight for each station or compartment.

k. For class 2 aircraft only, compartment centroids, locations, and distances from the reference datum. These shall be the values used for the compartment loading data.

NOTE: If small detail sketches and descriptive information can be used to advantage in presenting the information required for weighing purposes in 3.7.9.5.2c, g, h, and i above, or in reducing complexity in the aircraft diagram, these data may be included on the "Weighing instruction" page. Add suitable cross references of the aircraft diagram such as "See Weighing instructions on page _____, for jig point detail."

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3.7.9.5.3 Fuel and oil loading data. Moments shall be given for fuel and oil quantities in pounds in appropriate increments from zero usable fuel and oil to maximum quantity indicated on the gage dials. The "normal full" fuel capacity (based on 6.0 lb/gal for Specification MIL-F-5572 fuel and 6.5 lb/gal for Specification MIL-F-5624 fuel) shall be identified in the table and the following notes added to the table:

"Full tanks with Specification MIL-F-5572 gasoline at 6.0 lb/gal."

or

"Full tanks with Specification MIL-F-5624, JP-2 (JP-4) at 6.5 lb/gal."

or

"Full tanks with Specification MIL-F-5624, JP-5 at 6.8 lb/gal."

"Total weight of fuel is dependent upon the specific gravity and temperature. Therefore, the notation 'full' does not appear on the fuel quantity gages. Variation should be anticipated in gage readings when tanks are full."

When an aircraft can be operated with more than one type of fuel with varying specific gravity, tables shall be included for all types; if the effect on the moment data, because of varying fuel specific gravity, does not materially affect the aircraft c.g., the data may be combined into a single table using average data. The capacity of each tank as determined by actual calibration shall be stated. If it is necessary to use a calculated capacity for the initial issue of Chart E, the capacity listed shall be appropriately labeled, and actual data inserted in the earliest possible revised issue of Chart E. Dry weight, capacity, type, and moment data shall be given for all external droppable tanks, including any attachments not listed on Chart A. The fuel and oil c.g. used in computing the moments shall be tabulated. The effect on variation in fuel c.g. for varying amounts of fuel shall be considered in preparing the moment data. In such cases, the arms may be listed for 1/4, 1/2, 3/4 and full for each tank. When tanks are interconnected to fill and drain as a single tank, they shall be totalized and treated as a single tank. When an automatic feed and transfer system is used, data shall be presented in a manner which will make clear the sequence in which fuel is loaded into and consumed from the system to permit a loading check to be made.

3.7.9.5.4 Water injection fluid data. Weights, arms, and moments shall be listed in appropriate increments (gallons and pounds) of fluid. The weight per gallon of fluid shall be listed.

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FUEL TABLE

WEIGHT (POUNDS)	MOMENT / 1000 — INTERCONNECTED FUSELAGE TANKS 1102 GAL ARM - VARIES	WEIGHT (POUNDS)	MOMENT / 1000 — INTERCONNECTED FUSELAGE TANKS 1102 GAL ARM - VARIES	WEIGHT (POUNDS)	MOMENT / 1000 — INTERCONNECTED FUSELAGE TANKS 1102 GAL ARM - VARIES
60	11.3	2760	605.5	5460	1218.7
120	26.6	2820	619.3	5520	1232.1
180	40.6	2880	633.0	5580	1246.0
240	54.3	2940	646.8	5640	1259.4
300	68.0	3000	660.6	5700	1272.8
360	81.8	3060	674.4	5760	1286.2
420	95.5	3120	688.3	5820	1300.7
480	109.3	3180	701.8	5880	1313.6
540	123.1	3240	715.4	5940	1327.0
600	136.9	3300	729.3	6000	1340.4
660	148.8	3360	742.9	6060	1353.8
720	152.7	3420	756.5	6120	1367.2
780	166.5	3480	770.1	6180	1380.6
840	180.3	3540	783.8	6240	1394.0
900	194.0	3600	797.4	6300	1407.4
960	207.8	3660	811.1	6360	1421.5
1020	221.7	3720	824.7	6420	1434.9
1080	235.4	3780	838.0	6480	1448.3
1140	243.4	3840	851.7	6540	1462.3
1200	248.2	3900	865.4	*6600	1475.8
1260	261.0	3960	879.1	6660	1489.2
1320	275.9	4020	892.4	6720	1502.6
1380	289.7	4080	906.2	6780	1516.0
1440	303.4	4140	919.9	6840	1529.4
1500	317.1	4200	933.2	6900	1542.8
1560	330.7	4260	947.0	6960	1556.3
1620	344.4	4320	960.8	7020	1569.7
1680	357.5	4380	974.1	7080	1583.1
1740	371.3	4440	987.5	**7140	1596.5
1800	385.2	4500	1001.3	7200	1609.9
1860	399.0	4560	1015.1	7260	1623.3
1920	412.6	4620	1028.4	7320	1636.8
1980	426.5	4680	1042.2	7380	1650.7
2040	440.2	4740	1055.6	7440	1663.6
2100	453.8	4800	1069.4	7500	1677.0
2160	467.9	4860	1082.8	7560	1690.4
2220	481.7	4920	1096.7	7620	1703.8
2280	495.4	4980	1110.0	7680	1717.2
2340	509.2	5040	1123.4	7740	1730.7
2400	523.0	5100	1137.3		
2460	536.8	5160	1150.7		
2520	500.6	5220	1164.1		
2580	564.2	5280	1178.0		
2640	578.2	5340	1191.4		
2700	591.8	5400	1205.3		

WEIGHT (POUNDS)	MOMENT / 1000 — TIP TANK FUEL (2) 170 GAL Ea. ARM - VARIES
60	11.9
120	23.8
180	35.7
240	47.6
300	59.5
360	71.4
420	83.2
480	95.1
540	107.0
600	118.8
660	130.7
720	142.5
780	154.4
840	166.2
900	177.9
960	189.8
1020	201.6
1080	213.3
1140	225.2
1200	236.9
1260	248.6
1320	260.3
1380	272.1
1440	284.0
1500	295.7
1560	307.5
1620	319.3
1680	331.1
1740	341.0
1800	354.8
1860	366.6
1920	378.4
1980	390.5
2040	402.5
2100	414.3
2160	426.2
2220	438.0
2280	449.8
2340	461.7
2400	473.5

FUEL TANK SEQUENCE

INTERCONNECTING FUEL LINES OF THE THREE FUSELAGE TANKS MAKE IT POSSIBLE TO CONSIDER FUSELAGE FUEL AS BEING CONTAINED IN A SINGLE TANK UNTIL FUEL LEVEL REACHES W. L. 32.4.

DELETE ITEM 1 FROM FOLLOWING SEQUENCE WHEN TIP TANKS ARE NOT INSTALLED.

1. OPERATION OF WING TIP FUEL TRANSFER SWITCH TRANSFERS WING TIP FUEL TO CENTER FUSELAGE TANK, REPLACING FUEL CONSUMED.
2. FUEL IS CONSUMED FROM FORWARD, CENTER AND AFT FUSELAGE TANKS UNTIL FUEL LEVEL REACHES W. L. 32.4.
3. FUEL IS PUMPED FROM FORWARD AND AFT TANKS TO CENTER TANK UNTIL CENTER TANK FUEL LEVEL REACHES W. L. 47.0.
4. FUEL IS CONSUMED FROM CENTER TANK UNTIL CENTER TANK FUEL LEVEL AGAIN REACHES W. L. 32.4.
5. ABOVE SEQUENCE (ITEMS 3 & 4) CONTINUES UNTIL FORWARD AND AFT TANKS ARE EMPTY.
6. REMAINING FUEL IS CONSUMED FROM CENTER TANK.

NOTES

1. (*) SINGLE ASTERISK INDICATES APPROXIMATE WEIGHTS AND MOMENTS FOR FULL FUSELAGE TANKS BASED ON GASOLINE (MIL-F-5572) AT 6.0 LBS PER GALLON.
2. (**) DOUBLE ASTERISK INDICATES APPROXIMATE WEIGHTS AND MOMENTS FOR FULL FUSELAGE TANKS BASED ON JP-3 FUEL (MIL-F-5624) AT 6.5 LBS PER GALLON.
3. FUEL WEIGHT, IN FORWARD, CENTER AND AFT FUSELAGE TANKS AND TIP TANKS (WHEN INSTALLED), IS TOTALIZED ON ONE SCALE OF THE FUEL GAGE.

FUEL ARM TABLE

FUEL QUANTITY	ARM
1/4 FULL	212.7
1/2 FULL	220.7
3/4 FULL	222.8
FULL	223.6

TOTAL WEIGHT OF FUEL IS DEPENDENT UPON THE SPECIFIC GRAVITY AND TEMPERATURE. THEREFORE, THE NOTATION "FULL" DOES NOT APPEAR ON THE FUEL QUANTITY GAGES. VARIATION SHOULD BE ANTICIPATED IN GAGE READINGS WHEN TANKS ARE FULL.

FIGURE 10. Sample fuel loading data.

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3.7.9.5.5 Stores loading data. Moments shall be given for various weights of stores (bombs, clusters, special stores, torpedoes, depth bombs, etc.) for each bomb bay or station, together with the arms on which the moments are based. Maximum loadings shall be stated for each station. For cases where slight variations of arms occur because of various type and size store carried on a given rack or at a given station, an average arm chosen to yield the minimum moment error shall be used. Consideration shall be given to a separate tabulation for a given type or size store if the aircraft c.g. would be materially affected by including this type or size store at the average arm. Weights, arms, and moments, together with appropriate explanations, shall be included when special racks, pylons, or adapters are required which are not included on the Chart A.

3.7.9.5.6 Ammunition loading data. Ammunition weights, arms, and moments shall be given for each gun station, including the compartment designation, ammunition capacity, and caliber. For guns fed from a remote source, ammunition in feed chutes shall be listed separately, and capacity indicated. When ammunition cases and links are retained in the aircraft, a note to this effect (with weight, arm, and moment data for the retained items) shall be included. When ballast is provided, or required, for installation when the aircraft is flown without retained cases and links, weight, arm, and moment data, and identification of ballast by part number if contractor-furnished, shall be tabulated with appropriate notes concerning its use.

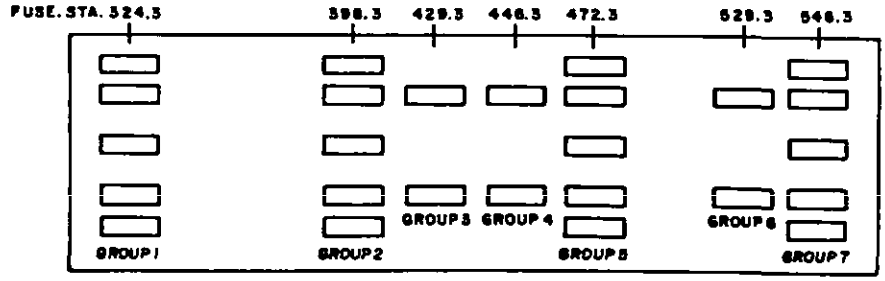
3.7.9.5.7 Rocket projectile loading data. Rocket weights, arms, and moments, including location, maximum quantity, and type shall be given.

3.7.9.5.8 Camera loading data for photographic aircraft. Weights, arms, and moments shall be given for cameras, magazines and film, flashbombs, and typical camera and magazine combinations. When special mounts or ballast, variable with the camera installation, are utilized, weights, arms, and moments shall be included for these items with appropriate explanations.

3.7.9.5.9 Assisted takeoff (ATO) data. Weights, arms, and moments shall be given for the ATO bottles in the "installed" location, including the compartment designation, number of units, and size. Data shall also be shown for the empty units with charge expended. If provision is made for stowing extra units, data shall also be listed for the units in "stowed" location and the data labeled "installed" and "stowed."

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BOMB STATIONS AND BOMB CARRIER POSITIONS



BOMB TABLE

	GROUP 1	GROUP 2	GROUP 3	GROUP 4	GROUP 5	GROUP 6	GROUP 7
ARM	324.3"	398.3"	429.3"	446.3"	472.3"	529.3"	546.3"
WEIGHT POUNDS	MOMENT/1000 FOR GROUP AT ARM NOTED						
25	8	10	11	11	12	13	14
50	16	20	21	22	24	26	27
75	24	30	32	33	35	40	41
100	32	40	43	45	47	53	55
200	65	80	86	89	94	106	109
300	97	119	129	134	142	159	164
400	130	159	172	179	189	212	219
500	162	199	215	223	236	265	273
600	195	239	258	268	283	318	328
700	227	279	301	312	331	371	382
800	259	319	343	357	378	423	437
900	292	358	386	402	425	476	492
1000	324	398	429	446	472	529	546
1100	357	438	472	491	520	582	601
1200	389	478	515	536	567	635	656
1300	422	518	558	580	614	688	710
1400	454	558	601	625	661	741	765
1500	486	597	644	669	708	794	819
1600	519	637	687	714	756	847	874
1700	551	677	730	759	803	900	929
1800	584	717	773	803	850	953	983
1900	616	757	816	848	897	1006	1038
2000	649	797	859	893	945	1059	1093
2100	681	836	902	937	992	1112	1147
2200	713	876	944	982	1039	1164	1202
2300	746	916	987	1026	1086	1217	1256
2400	778	956	1030	1071	1134	1270	1311
2500	811	996	1073	1116	1181	1323	1366
2600	843	1036	1116	1160	1228	1376	1420
2700	876	1075	1159	1205	1275	1429	1475
2800	908	1115	1202	1250	1322	1482	1530
2900	940	1155	1245	1294	1370	1535	1584
3000	973	1195	1288	1339	1417	1588	1639
3100	1005		1331	1384		1641	
3200	1038		1374	1428		1694	
3300	1070		1417	1473		1747	
3400	1103		1460	1517		1800	
3500	1135		1503	1562		1853	
3600	1167		1545	1607		1905	
3700	1200		1588	1651		1958	
3800	1232		1631	1696		2011	
3900	1265		1674	1741		2064	
4000	1297		1717	1785		2117	
4100				1830			
4200				1874			
4300				1919			
4400				1964			
4500				2008			

Aero 12A shackle, 16 lb each, used with 2,000-lb class stores.
Mk 7 Mod 1 shackle, 8 lb each, used with smaller class stores.

CAUTION: See arming and dropping sequencing instructions in Flight Handbook. To avoid extreme aircraft cg shift, stores must be loaded and dropped alternately from groups 1 and 7, and from groups 2 and 5.

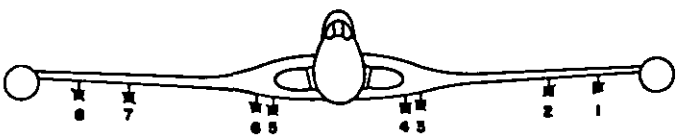
FIGURE 11. Sample stores loading data for class 2 aircraft.

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AMMUNITION LOADING TABLE

ARMAMENT 4 - 20 M M GUNS

ROUNDS PER GUN	WEIGHT PER GUN	MOMENT/1000		TOTAL AMMUNITION 4 GUNS ARM 67.3			TOTAL ESTIMATED CASES & LINKS ARM 64.0	
		INBOARD LH OR RH GUN ARM 62.0	OUTBOARD LH OR RH GUN ARM 72.5	ROUNDS	WEIGHT	MOMENT 1000	WEIGHT	MOMENT 1000
10	6.2	.4	.4	40	24.8	1.6	10.8	.7
20	12.4	.8	.9	80	49.6	3.4	21.6	1.4
30	18.6	1.2	1.3	120	74.4	5.0	32.4	2.1
40	24.8	1.5	1.8	160	99.2	6.6	43.2	2.8
50	31.0	1.9	2.2	200	124.0	8.2	54.0	3.5
60	37.2	2.3	2.7	240	148.8	10.0	64.8	4.1
70	43.4	2.7	3.1	280	173.6	11.6	75.6	4.6
80	49.6	3.1	3.6	320	198.4	13.4	86.4	5.5
90	55.8	3.5	4.0	360	223.2	15.0	97.2	6.2
100	62.0	3.8	4.5	400	248.0	16.6	108.0	6.9
110	68.2	4.2	4.9	440	272.8	18.2	118.8	7.6
120	74.4	4.6	5.4	480	297.6	20.0	129.6	8.3
130	80.6	5.0	5.8	520	322.4	21.6	140.4	9.0
140	86.8	5.4	6.3	560	347.2	23.4	151.2	9.7
150	93.0	5.8	6.7	600	372.0	25.0	162.0	10.4



BOMB AND ROCKET TABLE

ITEM	BOMB OR ROCKET STATIONS	WEIGHT OF PAIR	ARM	MOMENT 1000
H.V.A.R. 5 INCH ROCKETS	1 & 8	280	216.0	60
	2 & 7	280	228.1	64
	3 & 6	280	202.6	57
	4 & 5	280	202.6	57
100 LB G.P. BOMBS	1 & 8	235	214.6	51
	2 & 7	235	226.7	53
	3 & 6	235	201.2	47
	4 & 5	235	201.2	47
250 LB G.P. BOMBS	3 & 6	534	201.0	107
	4 & 5	534	201.0	107
500 LB G.P. BOMBS	4 & 5	1106	201.0	222

NOTES

1. AUTOMATIC FIRING SEQUENCE BY STATIONS IS AS FOLLOWS:
   SINGLE - 1, 7, 3, 5, 8, 2, 6, 4
   PAIRS - 1 & 8, 2 & 7, 3 & 6, 4 & 5
2. ABOVE SEQUENCE OF FIRING BOMBS & ROCKETS SHOULD BE FOLLOWED AS ARBITRARY MANUAL SELECTION MAY RESULT IN CENTER OR GRAVITY BALANCE CONDITIONS WHICH EXCEED THE RECOMMENDED LIMITS
3. BOMBS & ROCKETS MAY BE CARRIED AS FOLLOWS:
   500 LB BOMBS - STATIONS 4 & 5
   250 LB BOMBS - STATIONS 3, 4, 5, 6
   100 LB BOMBS - ALL STATIONS
   ROCKETS - ALL STATIONS
4. BRITKER BOMBS FOR ROCKETS ARE CARRIED AT STATIONS 3 & 6 WHEN 500 LB BOMBS ARE INSTALLED

FIGURE 12. Sample ammunition, rocket projectile, and stores loading data for class 1A and 1B aircraft.

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TABLE OF INDIVIDUAL CAMERA & EQUIPMENT WEIGHTS

ITEM	WEIGHT	ARM	MOMENT 1000	ITEM	WEIGHT	ARM	MOMENT 1000
FORWARD BAY				AFT BAY			
Cameras:				Cameras:			
S-7-S	69	38	2.6	K-18-24	66	58	3.8
K-17-6	37	39	1.4	K-17-24	59	58	3.4
K-17-12	43	39	1.7	K-17-6(3)	111	58	6.4
CA-8	40	39	1.6	Magazines:			
Magazines:				A-7	27	58	1.6
A-5A	29	39	1.1	A-8	44	58	2.6
A-9	37	39	1.4	A-5A(3)	87	58	5.0
MA-6	32	39	1.2	A-9	37	58	2.1
Amplifier	24	42	1.0	Ballast (PN 139949)*	118	12	1.4
				Ballast (PN 142432)**	175	38	6.7

TOTAL WEIGHTS AND MOMENTS OF TYPICAL CAMERA & MAGAZINE COMBINATIONS	AFT BAY - ARM = 58" (APPROX.)									
	CAMERA & MAGAZINE COMBINATIONS									
	K-18-24 & A-7		K-18-24 & A-8		K-17-24 & A-5A		K-17-24 & A-9		(3) K-17-6* & A-5A	
FWD. BAY - ARM = 39" (APPROX.)										
CAMERA & MAGAZINE COMBINATIONS	WEIGHT	MOM. 1000	WEIGHT	MOM. 1000	WEIGHT	MOM. 1000	WEIGHT	MOM. 1000	WEIGHT	MOM. 1000
S-7-S & Amplifier	186	9.0	203	10.0	181	8.7	189	9.1	291*	15.0
K-17-6 & A-5A	159	7.9	176	8.9	154	7.6	162	8.0	264*	13.9
K-17-6 & A-9	167	8.2	184	9.2	162	7.9	170	8.3	272*	14.2
K-17-12 & A-5A	165	8.2	182	9.2	160	7.9	168	8.3	270*	14.2
K-17-12 & A-9	173	8.5	190	9.5	168	8.2	176	8.6	278*	14.5
CA-8 & MA-6	165	8.2	182	9.2	160	7.9	168	8.3	270*	14.2

*Ballast (PN 139949) required in ALL camera combinations, except those marked with an asterisk (*).
**Ballast (PN 142432) required in lieu of cameras when NO cameras are aboard.

FIGURE 13. Sample camera data for photographic aircraft.

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3.7.9.5.10 Compartment loading data for class 2 aircraft

3.7.9.5.10.1 Cargo transport types. For all cargo compartments, moments shall be listed for cargo weight from zero to maximum capacity in appropriate increments based on the area centroid of the compartment. The capacity shall be considered to be the structural capacity (as limited by basic structural considerations), or the maximum allowable floor load, whichever is the lower. A supplementary table shall be furnished, listing for each compartment the compartment designation, centroid, arm limits, capacity, floor area, volume, maximum uniformly distributed load over the entire area (and over a limited area if variable), maximum concentrated load over a stated maximum area (such as floor beams, etc.). If the capacity is limited by floor strength and if the loading carrying capability of the aircraft might accommodate additional cargo weight, include a statement that the compartment capacities may be increased _____pounds by installation of supplementary or overlay flooring. In sectionalizing the cargo area of large cargo-type aircraft into compartments, the compartment lengths shall not exceed values obtained from the following parameters:

$$L = \frac{0.15\ cW}{G} \text{ for aircraft}$$

or

$$L = \frac{RW}{G} \text{ for rotorcraft}$$

Where L = maximum length in inches for equal length compartments
c = MAC length in inches for aircraft
R = Length of permissible c.g. range in inches for rotorcraft
G = Maximum takeoff gross weight minus (minimum operating basic weight plus weight of the specified crew), or 75 percent of the maximum possible cargo load based on fuselage structural limitations, whichever is the lower.
W = Minimum operating basic weight plus weight of specified crew plus G.

Individual compartment lengths may be increased or decreased to allow the compartment limits to fall at fuselage frames, discontinuities in floor plan or allowable floor loading, etc., or to avoid falling in between cargo or entrance door limits, etc., provided that after such adjustments:

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COMPARTMENT DATA

COMPARTMENT	A	B	C	D	E	F	G	H	I	J	K	L	M	N
FORWARD LIMIT	80	150	260	351	432	545	629	718	807	879	944	1007	1093	1172
AFT LIMIT	150	260	351	432	545	629	718	807	879	944	1007	1093	1172	1258
FLOOR AREA - SQUARE FEET			64	63	89	71	6	74	59	53	49	60	47	29
MAX. CAPACITY POUNDS PER SQUARE FOOT			45	45	45	45	45	45	45	45	45	45	45	45
** MAX. CAPACITY - POUNDS		800	2900	2800	4000	3200	3400	3300	2700	2400	2200	2700	2100	1300
* MAX. CAPACITY WITH CHART A ITEMS INSTALLED - POUNDS		149	2582	2409	2826	2195	1427	1622	2005	1328	1248	2404	1839	716

* Crew weight must be included in these capacities. Capacities for compartments E, K, and M do not include the weights of ammunition, sonobuoys and flares for which separate tables and capacities are listed on sheets 5 and 6 of Chart E.

** Includes the increase in maximum capacity possible by removal of equivalent weight of equipment listed on Chart A.

COMPARTMENT TABLE

COMPARTMENT		B	C	D	E	F	G	H	I	J	K	L	M	N
COMPARTMENT CENTROID		207	306	392	489	587	674	763	843	912	976	1050	1133	1215
	WEIGHT	MOMENT/1000 FOR COMPARTMENTS AT CG LISTED ABOVE												
	10	2	3	4	5	6	7	8	8	9	10	11	11	12
	20	4	6	8	10	12	13	15	17	18	20	21	23	24
	30	6	9	12	15	18	20	23	25	27	29	32	34	36
	40	8	12	16	20	23	27	31	34	36	39	42	45	49
	50	10	15	20	24	29	34	38	42	46	49	53	57	61
	60	12	18	24	29	35	40	46	51	55	59	63	68	73
	70	14	21	27	34	41	47	53	59	64	68	74	79	85
	80	17	24	31	39	47	54	61	67	73	78	84	91	97
	90	19	28	35	44	53	61	69	76	82	88	95	102	109
	100	21	31	39	49	59	67	76	84	91	98	105	113	122
	200	41	61	78	98	117	135	153	169	182	195	210	227	243
	300	62	92	118	147	176	202	229	253	274	293	315	340	365
	400	83	122	157	196	235	270	305	337	365	390	420	453	486
	500	104	153	196	245	294	337	382	422	456	488	525	567	608
	600	124	184	235	293	352	404	458	506	547	586	630	680	729
	700	145	214	274	342	411	472	534	590	638	683	735	793	851
	800	166	245	314	391	470	539	610	674	730	781	840	906	972
	900		275	353	440	528	607	687	759	821	878	945	1020	1094
	1000		306	392	489	587	674	763	843	912	976	1050	1133	1215
	1200		367	470	587	704	809	916	1012	1094	1171	1260	1360	1458
	1400		428	549	685	822	944	1068	1180	1277	1366	1470	1586	1701
	1600		490	627	782	939	1078	1221	1349	1459	1562	1680	1813	
	1800		551	706	880	1057	1213	1373	1517	1642	1757	1890	2039	
	2000		612	784	978	1174	1348.	1526	1686	1824	1952	2100	2266	
	2200		673	862	1076	1291	1483	1679	1855	2006	2147	2310	2493	
	2400		734	941	1174	1409	1618	1831	2023	2189		2520		
	2600		796	1019	1271	1526	1752	1984	2192			2730		
	2800		857	1098	1369	1644	1637	2136	2360			2940		
	3000		918		1467	1761	2022	2289						
	3200				1565	1878	2157	2442						
	3400				1663		2282	2594						
	3600				1760									
	3800				1858									
	4000				1956									

FIGURE 14. Sample compartment loading data for class 2 aircraft (for other than cargo transport types).

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$$g_1 L_1 + g_2 L_2 + \cdots + g_n L_n = GL \qquad (2)$$

where $L_1, L_2, \cdots L_n$ = individual compartment lengths

$g_1, g_2, \cdots g_n$ = individual compartment capacities when G is based on maximum takeoff gross weight, or 75 percent of individual compartment capacities when G is based on fuselage structural limitations.

For aircraft in which several cargo areas are provided, the cargo values for g and G in the above parameters shall be considered as the cargo in all such areas. For other than the cargo compartments, the requirements of 3.7.9.5.10.2 shall apply.

3.7.9.5.10.2 All other types. Moments shall be listed for miscellaneous loads, for which separate capacities are not listed elsewhere in Chart E, from zero to maximum capacity in appropriate increments. Miscellaneous loads will normally consist of crew, baggage, cargo, and items of an unspecified nature that might be stowed on ferry or special missions utilizing available floor area and stowage locker capacities. The compartment centroids shall be based on the weighed average arms of the miscellaneous loads. The capacity shall be considered to be the structural capacity (as limited by basic structural considerations), or a capacity based on summation of the known items of load plus allowable floor and stowage locker loads, whichever is the lower. A supplementary table shall be furnished, listing for each compartment the compartment designation, centroid, arm limits, capacity, floor area and maximum floor loading, and stowage locker capacities. A statement shall be included that crew weight must be included in these capacities, that the capacity does not include the weight of ammunition and other Chart E items for which separate tables and capacities are provided, and that capacities can be increased _____ pounds, where applicable, by the removal of an equivalent weight of equipment listed on Chart A or by installation of supplementary flooring.

3.7.9.5.10.3 Basis for compartment capacities centroids. The basis for the compartment capacities, the centroids chosen, and data substantiating the compartment lengths for cargo transport aircraft shall be furnished to the procuring activity with the sample Chart E.

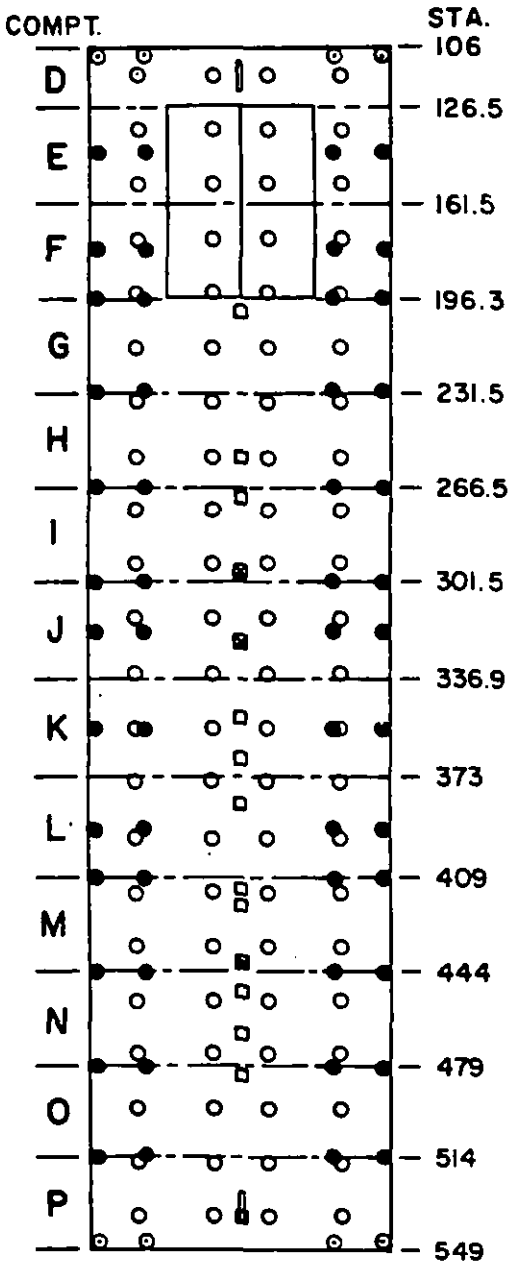
3.7.9.5.11 Cargo tiedown data for cargo types. A plan view of the cargo floor shall be provided showing the following:

a. Location and capacity of tiedown fittings. Compartment letter designations and station limits.

b. A note shall be added "Weight of cargo secured must not exceed fitting capacity. Refer to Cargo Loading Handbook for detailed information."

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CARGO TIEDOWN DATA



LEGEND

- ○ AN7516 FITTING
- ● 10,000 LB FITTING
- ⊙ 5,000 LB FITTING
- ■ ENGINE HOLD DOWN FITT'G 12,000 LB
- ◘ ENGINE HOLD DOWN FITT'G 7,000 LB

-ULTIMATE CAPACITIES-

NOTES:

(1) Weight of cargo secured must not exceed fitting capacities. Refer to Cargo Loading Handbook for detailed information.

(2) Compartment capacities must not be exceeded. See compartment capacities on sheet 4.

FIGURE 15. Sample cargo tiedown data for class 2 cargo types.

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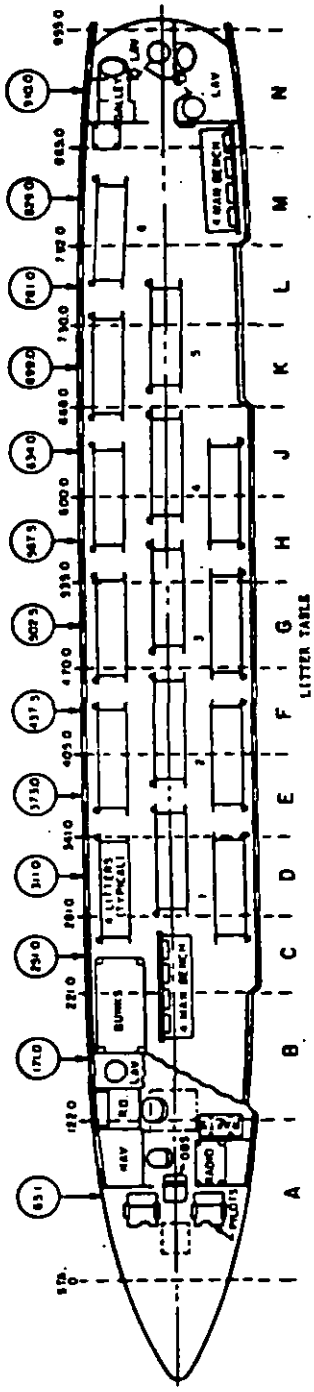
3.7.9.5.12 Passenger, troop, and litter data. Weights, arms, and moments shall be given for passengers, troops, litters, and patients, with each position, seat, row, etc., identified in the same manner as shown on the aircraft diagram.

3.7.9.5.13 Personnel movement data. For class 1A and 1B aircraft, moment arms and moments shall be given for each crew station, and moment changes accompanying shift of personnel to any station to which movement in flight is possible. For class 2 aircraft, similar data, based on compartment centroids, shall be given for compartment occupied or to which movement in flight is possible. Personnel weights shall be based on the weights specified by the procuring activity in the applicable design specifications. When fixed stowage is provided for parachutes, the personnel weights shall be decreased by the parachute weight as specified in the applicable design specifications, and a note added "Based on _____ pounds per man. Parachutes listed in Chart A." When no stowage is provided for parachutes, a note shall be added "Based on _____ pounds per man, including parachute." When personnel weights vary (e.g., different weights for crew and passengers), the heaviest weight shall be used.

3.7.9.5.14 Miscellaneous equipment data. Compartment designations, quantities, weights, arms, and moments shall be given for miscellaneous items of equipment not shown elsehwere in Chart E. Included in this category are heavy items of equipment, such as engines and cradles, tanks, jeeps, and field pieces for which specific stowage provisions are made; and flares, sonobuoys, etc., the weight and location of which materially affect aircraft c.g.

3.7.9.5.15 Center of gravity data. (Wherever "percent MAC" is referred to in this requirement, it shall be understood to apply to aircraft, and "inches aft of the reference datum" shall be understood to apply in the case of rotorcraft.) For various gross weights from basic to maximum, in equal weight increments based on the parameter given below, moments shall be tabulated in columns for the recommended forward and aft center of gravity limits and for several whole number percent MAC values spaced as evenly as possible between these limits. When special restrictions resulting from aerodynamic, structural, assisted takeoff, or hydrodynamic conditions are applicable for certain c.g. regions within the limits, moments for these limiting c.g. locations shall also be tabulated. Where appropriate, such special restrictions alternatively may be covered in note form, provided that the choice of percent MAC values for which moments are tabulated are carefully chosen and so spaced that a loading check for compliance with the restrictions may be made directly. If it is necessary to utilize a region at the aft end of the permissible c.g. range, in which the stability is not positive under all conditions of flight, this region shall be identified by "Caution See Note," and the following note included:

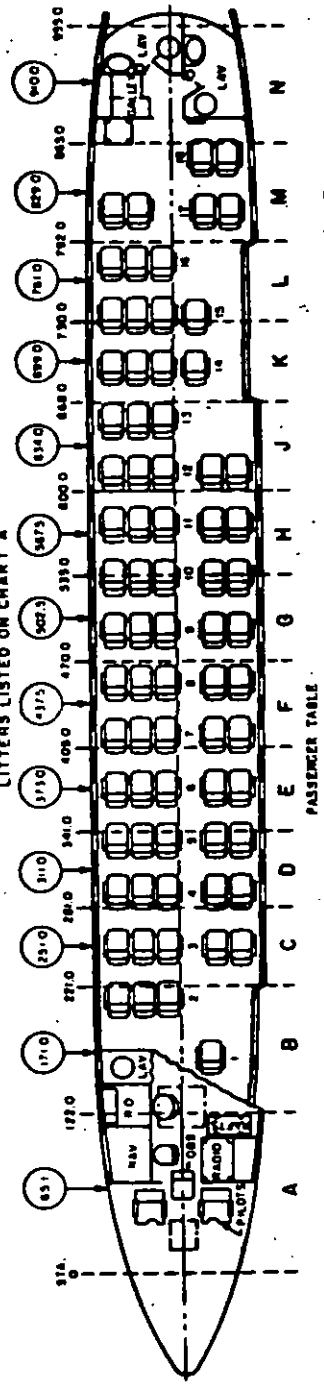
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LITTER TABLE

Location or Tier No.	Compt.	1 Pass /Compt. 170 lb. Mom/1000	2 Pass /Compt. 340 lb. Mom/1000	3 Pass /Compt. 510 lb. Mom/1000	4 Pass /Compt. 680 lb. Mom/1000	5 Pass /Compt. 850 lb. Mom/1000	6 Pass /Compt. 1020 lb. Mom/1000	7 Pass /Compt. 1190 lb. Mom/1000	8 Pass /Compt. 1360 lb. Mom/1000	9 Pass /Compt. 1530 lb. Mom/1000	10 Pass /Compt. 1700 lb. Mom/1000	11 Pass /Compt. 1870 lb. Mom/1000	12 Pass /Compt. 2040 lb. Mom/1000
Crew	A	11	22	33	44	-	-	-	-	-	-	-	-
Crew, Bench	B	29	58	87	116	145	-	-	-	-	-	-	-
Bench	C	43	85	-	-	-	317	370	423	476	529	582	634
	D	53	106	159	211	264	317	370	423	476	529	582	634
	E	63	127	190	254	317	380	443	506	569	634	698	761
	F	74	149	223	298	372	446	521	595	669	744	818	892
	G	85	171	256	342	427	513	598	683	769	854	940	1025
	J	108	216	323	431	539	647	754	862	970	1078	1186	1293
	K	119	238	356	475	594	713	832	951	-	-	-	-
5, Bench	H	141	282	423	564	705	846	987	-	-	-	-	-
Bench	N	135	-	-	-	-	-	-	-	-	-	-	-

BASED ON 170 LBS./PATIENT
LITTERS LISTED ON CHART A



PASSENGER TABLE

Location or Row No.	Compt.	1 Pass /Compt. 170 lb. Mom/1000	2 Pass /Compt. 340 lb. Mom/1000	3 Pass /Compt. 510 lb. Mom/1000	4 Pass /Compt. 680 lb. Mom/1000	5 Pass /Compt. 850 lb. Mom/1000	6 Pass /Compt. 1020 lb. Mom/1000	7 Pass /Compt. 1190 lb. Mom/1000	8 Pass /Compt. 1360 lb. Mom/1000	9 Pass /Compt. 1530 lb. Mom/1000	10 Pass /Compt. 1700 lb. Mom/1000
Crew	A	11	22	33	44	-	-	-	-	-	-
Crew, 1, 2	B	29	58	87	116	145	-	-	-	-	-
3	C	43	85	128	171	213	-	-	-	-	-
4, 5	D	53	106	159	211	264	317	370	423	476	529
6	E	63	127	190	254	317	-	-	-	-	-
7, 8	F	74	149	223	298	372	446	521	595	669	744
9	G	85	171	256	342	427	-	-	-	-	-
10, 11	H	96	193	289	386	482	579	675	772	868	965
12, 13	J	108	216	323	431	539	647	754	862	-	-
14	K	119	238	356	475	-	-	-	-	-	-
15, 16	L	129	259	388	517	647	776	906	-	-	-
17, 18	M	141	282	423	564	705	846	-	-	-	-

PARACHUTES FOR CREW ARE LISTED ON CHART A IN STOWED POSITION. OTHER PARACHUTES, WHEN CARRIED, SHALL BE CONSIDERED AS CARGO.

FIGURE 16. Sample of passenger and litter data.

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TABLE OF MOMENTS FOR PERSONNEL MOVEMENT (MOMENT/1000)

COMPARTMENT	B	C	D	E	F	G	H	I	J	K	L	M
ARM (INCHES)	207	306	392	489	587	674	763	843	912	976	1050	1133
ONE PASSENGER	37	55	71	88	106	121	137	152	164	176	189	204
N CABIN	182	164	148	131	113	98	82	67	55	43	30	15
M CABIN	167	149	133	116	98	83	67	52	40	28	15	
L CABIN	152	134	118	101	83	68	52	37	25	13		
K CABIN	139	121	105	88	70	55	39	24	12			
J CABIN	127	109	93	76	58	43	27	12				
I CABIN	115	97	81	64	46	31	15					
H CABIN	100	82	66	49	31	16						
G CABIN	84	66	50	33	15							
F CABIN	69	51	35	18								
E CABIN	51	33	17									
D CABIN	34	16										
C CABIN	18											

NOTE - ADD MOMENT FOR PASSENGER MOVEMENT AFT. PLUS (+) SIGN. SUBTRACT FOR MOVEMENT FORWARD, MINUS (-) SIGN.

Based on 180 lb per man. Parachutes listed on Chart A.

FIGURE 17. Sample personnel movement data for class 2 aircraft.

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"NOTE: Flight in the 'Caution' region shall be avoided if practicable, but the aircraft can be flown safely in this region provided that the pilot is familiar with its flight characteristics. Increasing caution should be observed as the c.g. approaches the Aft Limit."

The above note may be expanded to include a very brief statement of the flight conditions in which instability is encountered, and reference made to the applicable Technical Orders or Flight Handbooks discussing stability characteristics for the model, when such have been issued by the procuring activity.

3.7.9.5.15.1 Each column of the c.g. table shall bear an appropriate heading. The limit moment columns shall be identified by appropriate nomenclature, of which the statements listed below are considered representative, and the percent MAC and arm from the reference datum shall be listed. For limits which for any reason are tapered between two gross weights, the taper shall be identified. The following statements are representative of those considered satisfactory for c.g. limit nomenclature:

Fwd Limit Wheels down.
Fwd Limit Wheels up.
Fwd Hydro. Limit.
Aft Limit Wheels up.
Aft Hydro. Limit.

3.7.9.5.15.2 The weight increment used for the c.g. table shall be 5, 10, 25, 50, 100, or a whole number multiple of 100 pounds. The nearest increment of those listed above to the increment calculated from the following formula shall be used, except that when the value obtained from the formula is greater than 200 the next higher whole number multiple of 100 pounds may be used:

For aircraft $w = \frac{Wc}{100H}$

For rotorcraft $w = \frac{6RW}{100H}$

Where w = Weight increment
W = Minimum basic weight plus weight of specified crew
c = MAC in inches, for aircraft
H = Arm for aft c.g. limit in inches
R = Length of permissible c.g. range in inches, for rotorcraft.

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CENTER OF GRAVITY TABLE

(See sheet 7 for gross weights below 46,200 lb)
(See sheet 9 for gross weights above 57,000 lb)

CG LIMITS

GROSS WEIGHT (POUNDS)	MOMENT/1000									
	21% MAC (354.80)	22% MAC (357.86)	23% MAC (359.13)	24% MAC (360.39)	26% MAC (362.91)	28% MAC (366.44)	30% MAC (367.96)	32% MAC (370.48)	34% MAC (373.01)	36% MAC (375.53)
46,200	16475	16533	16592	16650	16766	16883	17000	17116	17233	17349
400	16546	16605	16664	16722	16839	16956	17073	17190	17308	17425
600	16618	16676	16735	16794	16912	17030	17147	17264	17382	17500
800	16689	16748	16807	16866	16984	17103	17221	17338	17457	17575
47,000	16760	16819	16879	16938	17057	17176	17294	17413	17531	17650
200	16832	16891	16951	17010	17129	17249	17368	17487	17606	17725
400	16903	16963	17023	17082	17202	17322	17441	17561	17681	17800
600	16974	17034	17095	17155	17275	17395	17515	17635	17755	17875
800	17045	17106	17166	17227	17347	17468	17588	17709	17830	17950
48,000	17117	17177	17238	17299	17420	17541	17662	17783	17904	18025
200	17188	17249	17310	17371	17492	17614	17736	17857	17979	18101
400	17259	17320	17382	17443	17565	17687	17809	17931	18054	18176
600	17331	17392	17454	17515	17637	17760	17883	18005	18128	18251
800	17402	17464	17526	17587	17710	17833	17956	18079	18203	18326
49,000	17473	17535	17597	17659	17783	17907	18030	18154	18277	18401
200	17545	17607	17669	17731	17855	17980	18104	18228	18352	18476
400	17616	17678	17741	17803	17928	18053	18177	18302	18427	18551
600	17687	17750	17813	17875	18000	18126	18251	18376	18501	18626
800	17759	17821	17885	17947	18073	18199	18324	18450	18576	18701
50,000	17830	17893	17957	18020	18146	18272	18398	18524	18651	18777
200	17901	17965	18028	18092	18218	18345	18472	18598	18725	18852
400		18036	18100	18164	18291	18418	18545	18672	18800	18927
600		18108	18172	18236	18363	18491	18619	18746	18874	19002
800		18179	18244	18308	18436	18564	18692	18820	18949	19077
51,000		18251	18316	18380	18508	18637	18766	18894	19024	19152
200		18322	18387	18452	18581	18711	18840	18969	19098	19227
400		18394	18459	18524	18654	18784	18913	19043	19173	19302
600		18466	18531	18596	18726	18857	18987	19117	19247	19377
800		18537	18603	18668	18799	18930	19060	19191	19322	19452
52,000		18609	18675	18740	18871	19003	19134	19265	19397	19528
200		18680	18747	18812	18944	19076	19208	19339	19471	19603
400		18752	18818	18884	19016	19149	19281	19413	19546	19678
600		18823	18890	18957	19089	19222	19355	19487	19620	19753
800		18895	18962	19029	19162	19295	19428	19561	19695	19828
53,000		18967	19034	19101	19234	19368	19502	19635	19770	19903
200		19038	19106	19173	19307	19441	19575	19710	19844	19978
400		19110	19178	19245	19379	19514	19649	19784	19919	20053
600		19181	19249	19317	19452	19588	19723	19858	19993	20128
800		19253	19321	19389	19525	19661	19796	19932	20068	20204
54,000		19324	19393	19461	19597	19734	19870	20006	20143	20279
200		19396	19465	19533	19670	19807	19943	20080	20217	20354
400		19468	19537	19605	19742	19880	20017	20154	20292	20429
600			19608	19677	19815	19953	20091	20228	20366	20504
800			19680	19749	19887	20026	20164	20302	20441	20579
55,000			19752	19821	19960	20099	20238	20376	20516	20654
200			19824	19894	20033	20172	20311	20450	20590	20729
400			19896	19966	20105	20245	20385	20525	20665	20804
600			19968	20038	20178	20318	20459	20599	20739	20879
800			20039	20110	20250	20392	20532	20673	20814	20955
56,000			20111	20182	20323	20465	20606	20747	20889	21030
200			20183	20254	20396	20538	20679	20821	20963	21105
400			20255	20326	20468	20611	20753	20895	21038	21180
600			20327	20398	20541	20684	20827	20969	21112	21255
800			20399	20470	20613	20757	20900	21043	21187	21330
57,000			20470	20542	20686	20830	20974	21117	21262	21405

CG LIMITS

The forward CG limit wheels down is 20% MAC for all weights below 46,000 pounds and tapers to 29% MAC wheels down at 54,000 pounds. The aft CG limit wheels up is 36% MAC. Limits are based on contractors tests. Limits determined during Navy tests will be furnished when available.

Moment/1,000 for retraction of the landing gear is minus 107. Loadings based on wheels down condition which fall within the limiting moments in the table will be satisfactory for flight with landing gear retracted.

GROSS WEIGHT LIMITATIONS

Takeoff ..... pounds*
Landing ..... pounds*
Zero wing fuel gross weight ..... pounds*

*Service activities shall insert, or substitute, current figures from the latest applicable Flight Handbook.

FIGURE 18. Sample center of gravity data.

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3.7.9.5.15.3 Since the Weight Handbook is predicated on use of basic weight, wheels down, the loading checks are made on the basis of the gross weight, wheels down, and the moment values for one of the c.g. limits may require compensation for the moment owing to retraction of the alighting gear. The applicable procedure of the following shall be used:

a. If retraction of the alighting gear shifts the horizontal c.g. aft, the moments listed for the aft limit (or limits, if special restrictions or a "caution" region are involved) shall be reduced numerically by the moment owing to retraction of the alighting gear and the following note added:

"NOTE: Moment/constant for retraction of the alighting gear is (plus) (fill in value). Moment/constant values listed in this column are the moments corresponding to (fill in) per cent MAC reduced by the moment for retraction of the alighting gear, and are equivalent to approximately (fill in) percent MAC at (fill in) pounds (contractor insert) figures for lowest weight of table) to (fill in) percent MAC at (fill in) pounds (contractor insert figures for highest weight of table). Loadings based on the wheels-down condition which fall within the limiting moments listed in the table will be satisfactory when the alighting gear is retracted."

b. If retraction of the alighting gear shifts the c.g. forward, and the forward limit is based on the flight requirements for a configuration in which wheels are normally retracted, the moments listed for the forward limit shall be numerically increased by the moment for retraction of the alighting gear, and the following note added:

"NOTE: Moment/constant for retraction of the alighting gear is (minus) (fill in value). Moment/constant values listed in this column are the moments corresponding to (fill in) per cent MAC increased by the moment for retraction of the alighting gear, and are equivalent to approximately (fill in) percent MAC at (fill in) pounds (contractor insert figures for lowest weight of table) to (fill in) percent MAC at (fill in) pounds (contractor insert figures for highest weight of table). Loadings based on the wheels-down condition which fall within the limiting moments in the table will be satisfactory when the alighting gear is retracted."

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c. If retraction of the alighting gear shifts the c.g. forward, and the forward limit is based on the requirements for landing or takeoff, during which the alighting gear is extended, the moments listed for the forward limit need not be compensated, and the following note shall be added:

"NOTE: Moment/constant for retraction of the alighting gear is (minus) (fill in value). Loadings based on the wheels-down condition which fall within the limiting moments in the table, will be satisfactory for flight with alighting gear retracted."

In the event the c.g. shift for retraction of the alighting gear is so large that the flight requirements discussed in (b) above become operative, the procedure in (b) shall apply.

3.7.9.5.15.4 When c.g. limits are based on contractor's data, the following note, modified to suit the particular case, shall be included:

"NOTE: C.g. limits based on contractor's tests (or calculations, if applicable). Limits determined during Navy (Air Force) tests will be furnished when available."

3.7.9.5.15.5 The following tabulation and note marked with an asterisk shall be added below (or on the last page) of the c.g. table, with blank space provided for insertion of weight figures by Service activities:

Gross weight limitations

Takeoff	_____	Pounds	*
Landing	_____	Pounds	*
Catapulting (when applicable)	_____	Pounds	*
Arresting (when applicable)	_____	Pounds	*

*NOTE: Service activities shall insert, or substitute, current figures from latest applicable (Technical Order) (Flight Handbook) covering operating restrictions.

For aircraft which have a gross weight restriction above which all weight must be fuel in the wings (e.g., a Zero Wing Fuel Gross Weight), list the "gross weight" properly identified.

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3.7.9.5.15.6 For aircraft with alternate alighting gear, the c.g. limit table shall include limits for the alternate alighting gear conditions. These may be separate tables if necessary for clarity.

3.7.9.5.15.7 When Weight and Balance Status Reports are not required, the contractor shall furnish to the procuring activity with the sample Chart E references to the source of the recommended c.g. limits, brief indication of the critical flying qualities or structural considerations which dictated these limits, and reasons for any tapered or special c.g. restrictions.

3.7.9.5.16 Maximum package size data (for cargo transport aircraft only). Tabular data shall be given for determining the maximum allowable cargo package sizes, in all practicable combinations of dimensions, which can pass through the cargo loading doors. A sketch shall be included identifying the door openings concerned and showing their dimensions.

3.7.9.5.17 Miscellaneous data. A table of miscellaneous general information shall present data as follows: Seat (including crew and passenger) and litter centroids, general aircraft dimensions, including span, length, height (maximum), approximate wheel base and tread.

3.7.9.5.18 Typical loadings. For class 1A and class 1B aircraft only, a table of several typical service loads shall be included listing combinations of Chart E items with weights, arms, and moments for each item. Totals of the weights and moments for the items in each typical loading shall be shown.

3.7.9.6 Approval of Charts A and E. The following procedures shall apply in initial preparation and subsequent revision of these charts:

a. Sample Charts A and E shall be submitted to the procuring activity for initial approval approximately 1 month before the expected flight date of the first aircraft.

b. When it is necessary to insert copies of Charts A and E in handbooks prior to release by the procuring activity, each page of the Charts A and E shall be marked "Temporary."

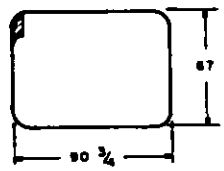
c. As soon as possible after approval, forward three prints of the released Charts A and E to the procuring activity. Text shall be clearly legible with no characters blurred or filled in, and no smaller than the equivalent of 6 point type.

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MAXIMUM PACKAGE SIZE TABLE

FORWARD CARGO DOOR

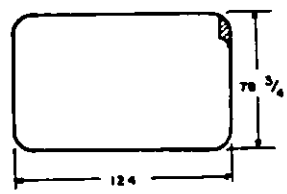
Width Inches	HEIGHT IN INCHES																		
	48 & Under	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66
	MAXIMUM LENGTH - INCHES																		
3	623	623	623	623	623	623	580	571	563	549	536	524	510	491	483	465	451	413	452
6	603	602	602	601	601	567	562	523	506	479	461	453	444	425	408	389	381	372	364
9	560	560	560	561	561	542	520	501	482	461	447	423	405	384	365	348	331	319	307
12	503	499	499	499	488	472	448	431	416	402	389	376	362	349	332	320	293	280	284
15	425	425	424	424	423	417	395	380	368	357	348	337	325	313	303	291	283	275	264
18	372	373	371	370	370	359	342	330	327	319	311	297	294	285	274	268	259	254	248
21	330	330	329	328	327	318	304	295	291	286	279	276	270	263	256	250	246	240	233
24	293	293	290	289	288	285	284	282	273	264	261	257	248	242	236	231	227	221	217
27	272	272	271	270	269	269	259	254	249	245	240	236	232	226	222	219	214	210	206
30	251	251	250	249	248	245	237	233	230	227	224	221	218	216	213	209	207	199	196
33	232	230	228	225	221	219	216	213	211	208	209	207	205	201	198	196	194	192	188
36	216	214	211	208	206	204	201	200	198	197	196	194	193	189	187	185	183	179	177
39	204	201	199	197	196	194	191	189	187	186	184	182	180	176	174	174	172	170	168
42	192	190	188	186	185	183	180	178	176	174	171	170	168	166	165	164	163	160	160
45	181	179	177	175	173	172	170	167	163	160	158	157	156	155	155	154	154	153	152
48	174	172	171	171	170	170	168	165	161	154	149	146	142	142	142	142	142	142	142
51	160	157	155	153	153	151	150	148	146	143	141	138	135	135	135	135	135	135	135
54	152	150	148	146	145	144	144	141	137	133	131	129	127	127	127	127	127	127	127
57	143	143	141	139	138	137	137	135	130	125	123	120	118	118	118	118	118	118	118
60	138	136	134	133	132	131	131	127	124	120	118	116	111	111	111	111	111	111	111
63	131	129	128	127	126	125	122	118	116	114	112	107	104	104	104	104	104	104	104
66	125	121	120	119	118	115	74	74	74	74	74	74	74	70	70	70	70	70	70
69	120	118	113	110	105	100	74	74	74	74	74	74	74	70	70	70	70	70	70
72	116	113	110	105	101	97	74	74	74	74	74	74	74	70	70	70	70	70	70



AFT CARGO DOOR

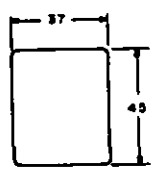
Width Inches	HEIGHT IN INCHES																		
	60 & Under	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78
	MAXIMUM LENGTH - INCHES																		
3	623	623	623	623	623	623	623	623	620	612	596	578	547	523	499	473	446	429	417
6	623	623	623	623	623	623	623	620	605	593	581	562	521	503	476	449	431	418	393
9	623	623	623	623	623	623	625	618	595	579	554	533	494	477	453	430	417	394	382
12	602	601	580	575	560	543	530	511	504	496	472	458	432	417	404	387	372	363	354
15	538	534	519	506	498	489	475	460	443	428	428	403	392	380	369	350	349	341	330
18	484	477	463	452	444	435	424	414	406	398	384	373	361	349	341	330	331	312	302
21	430	421	414	409	401	393	385	376	367	358	350	339	328	321	313	304	298	289	282
24	388	390	385	379	368	363	350	340	339	330	326	322	319	308	299	291	284	279	271
27	343	350	350	347	341	336	329	322	315	307	298	289	281	270	272	269	266	263	259
30	336	330	328	324	319	311	304	300	294	286	279	271	264	260	257	254	251	247	245
33	318	311	307	301	298	292	287	282	278	271	263	255	248	244	239	235	231	237	234
36	290	286	284	282	280	276	272	267	263	256	249	243	235	231	228	224	219	216	213
39	274	271	269	267	264	260	256	251	248	242	237	230	224	219	215	213	205	201	198
42	259	256	254	252	250	247	243	238	235	230	225	219	214	210	204	199	196	193	189
45	247	244	242	240	238	235	231	226	223	218	213	208	204	201	197	193	189	185	181
48	234	232	230	228	226	223	218	214	211	206	204	199	196	192	188	185	181	177	175
51	227	229	218	216	214	210	207	203	200	193	191	188	184	180	177	174	173	170	168
54	218	208	206	205	202	198	195	192	191	188	185	181	179	173	171	168	165	163	162
57	198	196	194	192	190	187	185	183	182	178	175	173	171	167	164	162	159	157	155
60	189	186	184	182	180	178	176	174	173	171	170	169	168	163	160	158	155	153	149
63	181	179	177	175	173	170	168	166	165	163	160	158	156	154	151	149	146	144	142
66	174	171	169	167	165	164	162	160	159	156	154	152	151	149	146	144	142	140	138
69	169	166	163	161	160	159	158	156	152	150	148	147	146	144	141	139	137	135	134
72	163	161	158	156	153	151	150	148	146	145	143	142	141	139	137	136	134	132	130
75	157	155	152	150	147	143	144	142	140	139	138	137	135	134	132	130	128	126	
78	151	149	146	144	141	139	138	136	134	133	132	131	130	128	126	125	123	118	
81	146	143	140	138	135	133	131	130	129	128	127	126	125	122	120	116			
84	141	138	135	132	129	128	127	126	125	124	123	122	120	118	115				
87	136	133	130	127	124	122	121	119	118	117	115	114	113						
90	132	128	125	122	121	120	118	117	116	115	113								
93	127	124	122	119	116	115	114	113	112	111									
96	124	121	119	116	114	113	113												
99	120	118	116	114	112														
102	117	115																	
105	115																		

SHADED PART SHOWS APPROX. AREA OBSTRUCTED DUE TO DOOR OPENING LINKAGE.



BELLY CARGO DOORS

Width Inches	HEIGHT IN INCHES																		
	8	9	12	15	16	17	18	19	20	21	22	23	24	25	26	27			
	MAXIMUM LENGTH - INCHES																		
3	168	166	160	160	160	160	150	149	142	137	131	126	120	113	104	101			
6	166	160	160	160	160	155	147	141	135	128	121	116	113	105	97	91			
9	166	160	160	150	144	139	135	128	123	117	113	108	104	96	89	82			
12	160	160	158	141	135	129	124	118	113	109	106	101	97	92	81	74			
15	166	160	142	127	122	118	114	109	104	101	98	94	90	79	72	67			
18	150	135	123	114	111	106	105	101	97	93	91	87	84	77	69	60			
21	126	116	106	99	96	95	94	90	87	84	82	79	77	68	61	54			
24	112	104	96	91	88	86	84	81	79	76	74	71	70	61	54	48			
27	99	95	89	82	79	78	77	74	72	70	68	66	64	53	49	43			
30	88	86	80	74	72	72	71	69	67	65	63	61	58	49	43	39			
33	80	78	73	70	68	67	66	64	63	60	57	54	53	45	41	36			
36	73	71	69	66	64	62	61	59	58	56	53	50	48	42	38	34			



NOTE: Maximum length of forward compartment - 130 inches. Packages longer than 130 in. can be loaded in the aft compartment only. This is indicated by dotted line.

FIGURE 19. Sample maximum package size data (for cargo transport aircraft only).

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MISCELLANEOUS DATA

CREW, PASSENGER AND LITTER CENTROIDS

COMPT	CREW		PASSENGER		LITTER	
	Crew Member	Arm	Row No.	Arm	Tier No.	Arm
B	Pilot & Co-Pilot Flight Engineer Radio Operator	190 222 226				
D	Navigator Relief Crew	308 329				
E	Relief Crew	407	1 2	390 427	1 2 3	428 428 448
F			3 4	464 501		
G			5 6	538 574	4 5 6	525 526 539
H			7 8	611 648	7 8 9	620 623 636
I			9 10	685 722	10	715
J			11 12	758 795	11 12	736 769
K			13 14	832 869	13 14 15	810 834 861
L			15 16	906 942	16 17	905 932
M			17 18	976 1016	18 19 20	955 1000 1029
N	Stewards or Attendants	1096 1134	19 20 21	1053 1090 1126		

DIMENSIONAL DATA

Span	117 ft
Length	115 ft 8 in.
Maximum height over	
fin - static altitude	30 ft 2 in.
- level altitude	32 ft 5 in.
Approx. wheel base	42 ft 5 in.
Approx. tread	28 ft 3 in.

FIGURE 20. *Sample miscellaneous data.*

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TYPICAL SERVICE LOAD CONDITIONS													
ITEM	ARM	NORMAL		OVERLOAD		COMBAT		FIGHTER BOMBER		ROCKET FIGHTER		DELIVERY	
		WEIGHT	MOM 1000	WEIGHT	MOM 1000	WEIGHT	MOM 1000	WEIGHT	MOM 1000	WEIGHT	MOM 1000	WEIGHT	MOM 1000
Pilot and Chute	140.0	200	28	200	28	200	28	200	28	200	28	200	28
Parakit and Seat Pan	142.1	26	4	26	4	26	4	26	4	26	4	26	4
Fuel (at 6 lb/gal)													
Main .....683 Gal	209.6	4098	859	4098	859			4098	859	4098	859	4098	859
Main .....554 Gal	209.0					3324	695						
Tip Tank.....240 Gal	230.1			1440	331			1440	331	1440	331		
Water (at 7.64 lb/gal). 22.5 Gal	268.0			172	46			172	46	172	46		
Ammunition - 20mm													
Inboard Guns...380 Rd	68.6	236	16	236	16	236	16	236	16	236	16		
Outboard Guns .380 Rd	139.7	236	33	236	33	236	33	236	33	236	33		
Ballast (in lieu of ammunition)	60.2											251	17
Rockets										840	183		
6 - 5 In. HVAR	218.2									2000	446		
2 - 6 Shot 5 In. HPAG Packages	223.0												
Bombs								1512	328				
6 - 250 lb G.P.	217.1							1090	243				
2 - 500 lb G.P.	223.0							47	11	47	11		
Sway Braces - Aero -1A	223.8												
Totals		4796	940	6408	1317	4022	776	9057	1899	9295	1957	4575	908

FIGURE 21. Sample typical loading for class 1A and 1B aircraft.

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d. When minor changes in either Chart A or E and reprinting becomes necessary subsequent to initial approval, the contractor shall redate the page(s) of the chart(s) affected and change the Chart Date on page 1 to agree. A copy of all revised pages plus three prints of the complete chart(s) shall be forwarded to the procuring activity as soon as possible, but not later than 1 week after the date of actual insertion in handbooks of the aircraft affected. A list of differences over the previous issue of the charts, the reason therefore, a summary of the latest revision dates of the pages which comprise the chart(s), the serial number of the first aircraft affected, and when only one chart is affected, the "chart date" of the other remaining in effect, shall also be furnished. The revised charts may be considered satisfactory unless otherwise advised by the procuring activity.

e. If important or extensive corrections of the Charts A and E, or major changes to the aircraft, warrant replacement of pages of the Charts A and E (or the complete Chart E, if appropriate) in the Weight Handbooks of aircraft already delivered, the procedure outlined in paragraph d above shall be applied and printed copies also furnished as indicated below:

(1) When c.g. limits based on procuring activity tests are furnished to supersede limits based on contractor tests or calculations, copies of the revised Chart E c.g. data shall be furnished to the local representative of the procuring activity for distribution directly to all operating activities affected, with appropriate instructions to the operating activities concerning the replacement of Chart E in the Weight Handbooks.

(2) When contractor-furnished service bulletins and parts kits involve major changes to be incorporated by operating activities and necessitate extensive corrections or additions to the Charts A and E (such as new fuel or ammunition tables for added or relocated tanks or guns, new compartment tables for increased floor strength, etc.), copies of the revised Charts A and E data shall be included in the parts kits, and appropriate instructions concerning the replacement of Charts A and E in the Weight Handbook included in the instructions for accomplishing the change.

(3) When errors are detected in the Charts A and E which materially affect the accuracy of the weight and c.g. calculations, the procedure of (1) above shall be applied.

f. When major changes involving a change in model designation are concerned, the procedure for initial approval specified in paragraphs a, b, c shall apply.

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3.7.10 Balance computer design data. For class 2 aircraft for which a balance computer is to be furnished, the procuring activity (if the computer is GFAE) or the aircraft contractor (if the computer is CPE) will forward one copy of the approved Charts A and E to the balance computer manufacturer for use as design data. The following information shall be furnished in duplicate with the charts:

a. Anticipated minimum and maximum basic weights, and most forward and aft basic weight c.g. locations, assuming removal of equipment for a minimum stripped condition and addition of specified special equipment for a maximum equipped condition.

b. When fuel and oil moment arms vary with fuel quantity, a table or graph of the variation in arm from empty to full tank(s) upon which the Chart E fuel and oil tables are based.

3.7.10.1 Release of balance computer designs. A sample faceplate to the proposed balance computer will be furnished to the procuring activity as well as to the aircraft contractor, who shall furnish comments to the procuring activity in duplicate as soon as possible on the completeness and conformance of the sample design with the balance computer design data, and any other subsequently available information on the aircraft. Comments of the contractor and the procuring activity will be coordinated and forwarded to the balance computer manufacturer by the procuring activity. In the event that the comments require revision of the design, samples and further comment shall be handled as above, except that when minor changes are involved, comments and final release will be handled by the procuring activity.

3.8 Balance calculations.

3.8.1 General. Balance calculations, as required for the respective weight reports, shall be prepared in tabular form with groups listed in the order shown on the weight statements (MIL-STD-1374, Parts I and II). Weights, arms, and moments shall be given in detail, with totals for each subgroup and group. Equipment groups (e.g., Electrical, Armament, Air Conditioning, etc.) shall be broken down to list "installations" and major components comprising these installations. Each engine, propeller, tank, etc., shall be listed separately, unless the same moment arm applies. The horizontal and vertical moment axes shall be so located that all moments will be positive. The axes shall be dimensioned from convenient fixed points on the body. Arms and moments for all weight report and the Weight Handbook shall be in inches and pound-inches, except that subject to prior approval by the procuring activity feet and pound-feet may be used for extremely large aircraft.

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3.8.2 Lateral center of gravity. These data are required unless specifically waived by the procuring activity. Detailed calculations shall be included in the Detail Weight Reports (Estimated, Calculated, First, and Last Aircraft) to show the lateral unbalance and the loading limitations or cautions necessary to control lateral distribution of load. A scale for determining lateral dimensions shall be included on the appropriate structural diagrams in the weight reports. The requirements of paragraphs 3.7.1b, 3.7.9.5.15 and 3.8 shall apply to lateral center of gravity limits, lateral center of gravity data and lateral balance calculations respectively.

3.8.3 Most forward and most aft center of gravity conditions. For classes 1A and 1B aircraft, these calculations are intended to determine the extreme c.g. positions attainable. For class 2 aircraft, these calculations are intended to determine the extreme c.g. positions which may occur using reasonable care in crew shifting, bomb drop sequence, fuel management, cargo, placement, etc. Such calculations will demonstrate that the c.g. can be maintained within the permissible c.g. range, and also will serve the purpose of determining what "cautions," "special instructions," etc., may be desirable to insure that operating activities know how to achieve proper balance or how to avoid the critical conditions. The most forward and most aft calculations shall show the extreme c.g. positions which may result from each of the following:

a. Any specified load condition, considering partial or complete expenditure of any expendable useful load items and necessary crew movement from normal flight stations to takeoff, landing, battle, or ditching stations.

b. The extremes of the above plus the most critical combination of special or alternate load items.

c. For personnel transport aircraft, assume full row seating in the most adverse manner to give extreme c.g. attainable. If these are outside the permissible c.g. range, show seating restrictions required. Where baggage is included and alleviates the critical loading, the condition of "no baggage carried" shall also be shown.

d. For cargo aircraft, determine most adverse distribution of cargo that can be carried and still maintain loadings within the permissible c.g. range. The calculations will be indicative of the degree of loading control needed within the load-carrying capacity of the aircraft, and demonstrate that, with rational distribution of cargo, the permissible cargo loads can be utilized within the permissible aircraft c.g. range.

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3.8.4 Alighting gear. Weight empty and basic weight c.g. locations shall be calculated in weight reports with alighting gear extended. Gross weight c.g. locations shall be calculated in weight reports with alighting gear extended and retracted. For calculations deriving the moment change for retraction of the alighting gear, moving and stationary items shall be listed separately.

3.8.5 Armament variables. For guns fed from a remote source, ammunition in feed chutes shall be listed separately. In the extreme balance calculations, the effect of empty cases and links shall be included when provision is made for retaining them in the aircraft. Partial loading or expenditure of stores, rockets, etc., in the most critical sequence, or in accordance with the operating sequencing instructions to be prescribed in the Flight Handbook, shall be considered when these items can be carried in more than one bay or on several stations.

3.8.6 Fuel variables. Quantities in each tank shall be listed as separate items unless the same moment arm applies. When tanks are interconnected to fill and drain as a single tank, weights, arms, and moments shall be listed for each tank and for the total quantity in the interconnected tanks. When more than one type of fuel is provided for, supplementary calculations based on nominal fuel weights shall be included for the alternate fuels. If variation in fuel density materially affects the extreme c.g. locations attainable, the effect of a density variation of ±0.6 pound/gallon from the nominal weight shall also be shown. If an extreme c.g. position will result with fuel tanks approaching an empty condition, the fuel shall be considered as totally consumed. Where emergency manual selection of tanks is provided on automatic sequencing systems, the extremes that could be encountered by adverse emergency selection shall also be indicated. The following information, as appropriate, shall be included in the Detail Weight Reports (Estimated, Calculated, First and Last Actual) where necessary to explain the fuel filling and consumption sequence and to substantiate the distribution assumed for the gross weight and extreme balance conditions:

a. Reason for or justification of the distribution assumed in the calculations, if the distribution is not automatically controlled and is not obviously the most critical attainable. This may consist of a statement of the filling and consumption sequences to be prescribed in the fuel system management instructions in the Flight Handbook, or that fuel usage must be controlled by pilot for balance control.

b. Brief description of system operation, if automatic distribution by single point fueling or automatic consumption sequence is provided. Include schematic diagrams outlining fuel system arrangement, brief explanation of distribution and sequencing features, and graphs showing distribution as system is filled and fuel is consumed. Where extreme attitudes in prolonged climb and descent materially affect the distribution, include sufficient information on the effect of attitude to permit evaluation of the attainable extreme aircraft c.g. location.

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3.8.7 Oil variables. When multitank or automatic distribution, transfer or feed system are provided, the applicable requirements of 3.8.6 shall be applied. Oil consumption may be neglected in extreme balance calculations unless total oil consumption will affect the attainable extreme c.g. locations by more than 0.5 percent MAC for aircraft, or 0.5 inch for rotorcraft. Rational assumptions of rate of consumption shall be made when it is necessary to consider expenditure of oil.

3.8.8 Miscellaneous variables. If "placards," which restrict the installation or combination of items, are to be installed in the aircraft, the proposed or actual location and wording shall be stated in the weight reports and the attainable extreme c.g. conditions may be based on assuming compliance with these placards. Partial installation of de-icer boots on wings, tail, and other surfaces need not be considered.

3.8.9 Tank capacities. When calibration of the fuel tanks has been completed, loading conditions which are based on full tank(s) shall be adjusted in the next weight report furnished, based on the actual capacities, and footnotes to this effect shall be made on the weight report. If the difference over the estimated fuel capacity is large, oil shall be adjusted based on the specified fuel/oil ratio. When calibration of the oil tank(s) has been completed, any difference between the estimated and actual capacity shall be noted specifically in the next weight report furnished.

3.9 Weighings.

3.9.1 General. Weighings shall provide for the determination of the weight empty, except that for instrumented flight test aircraft, weighings may determine as-delivered weight. It is preferable to make all weighings with the aircraft completely assembled and in the dry condition (before the fuel and oil systems have been filled). However, the first aircraft to be weighed on the contract shall be weighed in the dry condition. Aircraft with alternate alighting gear may be weighed in any one of the alternate alighting gear conditions. Weights shall be determined by taking the average of three independent readings (e.g. for beam scales by upsetting the beams of all scales between readings, or completely unloading the electronic load cells). Approval of the weighing procedures must be obtained from the procuring activity prior to the weighing.

3.9.2 Scales. The scales are to be calibrated or certified correct by Government or commercial scale officials satisfactory to the contractor and the Government Inspector at least once every year. Scale calibration periods may be modified as specifically requested by the procuring activity.

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3.9.3 Center of gravity determination. Weighings shall provide horizontal c.g. locations as required on the approved weighing forms. The vertical c.g. locations shall be determined by calculation. When specifically requested by the procuring activity, the vertical c.g. shall be determined as required by Form AN-9250. Lateral c.g. locations shall be determined when specifically requested by the procuring activity.

3.9.4 Trapped fuel and oil. Trapped fuel and trapped oil are defined as the amounts of fuel and oil remaining in the respective systems when oil is drained and the aircraft is defueled in static, ground attitude (seaplane on beaching gear) by the normal means provided for draining the tanks. The data are intended to establish a baseline for reweighing of aircraft and determination of weight empty or basic weight. Unless otherwise specified, the weights of trapped fuel and oil shall be determined on the first aircraft, and on a representative aircraft each time a change is made in the fuel and oil systems which will cause an appreciable change in the trapped fuel and oil data. The data shall be obtained in the manner indicated below, starting with an aircraft with compeltely dry systems (i.e., before any fuel or oil is placed in tanks). For aircraft with alternate alighting gears, trapped fuel and oil for the alternate alighting gear conditions shall be furnished; the data for the conditions not determined by actual weighing may be based on analysis and calculations to account for any material differences caused by attitude, float tanks, outlet locations,etc. Diagrams or sketches shall be included to identify specifically the drain points and their physical location. The distribution of the trapped quantities shall be rationally determined by calculating or estimating the quantities in the major components of the system (engine, oil coolers, tank sumps, etc.) to determine their weights, arms, and moments. These data shall be tabulated to show the derivation of the weights, arms, and moments for the totals of trapped fuel and trapped oil.

a. Trapped fuel:

(1) Add a measured weight of fuel into each tank. Where single point fueling is provided, insure that sufficient fuel is added to permit circulation to all tanks and plumbing.

(2) Operate fuel pumps to insure filling of all fuel lines. Do not operate the engine.

(3) Using only the normal means provided for defueling the aircraft, drain the fuel into containers. Weigh the containers before and after draining. Aircraft which have transfer tanks or separate long-range rank systems may in some cases not drain as completely in ground attitude as in flight. Where the pattern of previous missions and subsequent defueling would result in material differences in trapped fuel, include a determination of this difference with appropriate explanation.

(4) The trapped fuel is then the difference between the weights of fuel measured in (1) and (3).

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b. Trapped oil:

(1) Determine the weight of engine preoiling oil. Pour a measured weight of oil into each tank.

(2) Operate the engine the minimum length of time necessary to insure that the oil has circulated.

(3) Using only the drains provided for the oil tanks (usually the sump drains), drain the oil into a container. Weigh the container before and after draining.

(4) The trapped oil is then the difference between the weights of oil measured in (1) and (3).

3.10 Measuring and leveling provisions.

3.10.1 Application. The following paragraphs cover the general requirements for measuring and leveling provisions for use when weighing aircraft, when aligning fixed guns, or when structural alignment is being made on aircraft.

3.10.2 Measuring provisions. Each aircraft shall be provided with jig points for taking measurements during weighing and alignment. Each jig point shall consist of a hole or fitting, located such that it is a known distance $\pm$ 0.06 inches from all reference datums for the aircraft concerned. These measuring provisions shall cause no increase in drag when not in use. Each jig point shall be appropriately labeled so that its position on the aircraft may readily be determined. As a minimum, jig points shall be on the fuselage and wings if the aircraft has such. Jig points not on the centerline of the aircraft shall be symmetrically located on each side of the aircraft. The fuselage shall have jig points forward and aft of the wing. Each jig point shall be located such that nothing will interfere with dropping a plumb line to the hangar floor.

3.10.3 Leveling provisions. Each aircraft shall be furnished with the provisions specified herein for obtaining the true level attitude of the longitudinal and lateral axes of the aircraft while the aircraft is being weighed, while structural alignment is being made, or while fixed guns are being aligned. When provisions for leveling the aircraft or weighing or for structural alignment are not suitable for use in aligning fixed guns, a second set of leveling provisions shall be furnished for that purpose and marked "GUN LEVEL". All leveling provisions shall be clearly listed.

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3.10.3.1 Plumb bob. Provisions shall be made, as further specified below, for a plumb line swinging at a radius of not less than two feet within the aircraft, and in such a position that the relation between the plumb line or the point of the plumb bob and a graduate plate can be seen closely and without personnel entering or climbing on the aircraft. Subject to specific approval of the Procuring Agency, leveling personnel will be permitted aboard extremely large aircraft, the weights of which are such that leveling accuracy will not be impaired by minor movements of leveling personnel. In this latter case, provisions for leveling shall be readily accessible, shall be as near the center of gravity as practicable.

3.10.3.1.1 Upper bracket. An upper jig-located bracket shall be attached to structural members of the aircraft in such a position that a plumb line can be suspended therefrom. The bracket shall be designed in such a manner that the plumb line will always be suspended from the same point, and will be positioned with unvarying accuracy as related to the leveling plate or plates located below the bracket.

3.10.3.1.2 Lower plate. A metal plate or plates shall be permanently attached to some structural part of the aircraft below the upper bracket specified above. The lower plate or plates shall be positioned so that when a plumb line is suspended from the upper bracket either a portion of the plumb line or the point of the plumb bob will fall sufficiently close to the graduations on the plate or plates to permit easy reading of the indicated attitude. The graduations in degrees and minutes shall be permanently inscribed on the plates and shall provide for indicating the following aircraft attitudes:

a. Longitudinally and laterally level

b. Normal static ground attitudes between three degrees forward of the most nose down attitude to three degrees aft of the most tail down attitude, and laterally level

c. The attitudes for structural alignment and gun alignment if different from paragraph (a) above.

3.10.3.2 Leveling lugs. For aircraft not sufficiently large for use of the leveling aids previously specified, other leveling provisions as specified below shall be installed.

3.10.3.2.1 Two jig-located fittings shall be placed 10 to 20 inches apart on some primary structure of the aircraft. The fittings shall provide an ample firm support upon which to place a spirit level, and shall be so located as to give a true level reading and be accessible for leveling the aircraft. Similar provisions shall be made for laterally leveling the aircraft, preferably near the location of the longitudinal leveling fittings. All fittings shall be located so that personnel can take leveling readings without getting into or climbing on the aircraft, and may be located within the wheel well, door entrance or similar enclosures, or external to the aircraft.

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3.11 Weights.

3.11.1 GFAE weight. The contract/guaranteed weight empty is based on the weights of the GFAE listed in the contract detail specification as modified by specified weights of GFAE added by contract changes. Any overweight or underweight owing to addition, deletion, variation, or change of a Government-furnished item for which allowance has not been made in a contract change shall be considered as revising the contract/guaranteed weight empty accordingly.

3.11.2 Overweight and underweight.

a. Preproduction contracts: Unless otherwise specified, for preproduction contracts, the overweight and underweight shall be based on the actual weight, as determined by the weighings made upon completion of the aircraft, plus the weight incident to the incorporation if all changes and modifications (actually installed) found necessary to fulfill the requirements of the specifications and the amended contract for acceptance of the aircraft. The contractor shall establish an appropriate system for the recording of all modifications made subsequent to the weighings, together with their effect on weight and balance; this shall cover the periods during preliminary flights, procuring activity tests, etc., up to the time when the aircraft is finally accepted. The weight empty given in the original weight report and revised by the data in the appendices shall serve as the basis for the determination of the overweight or underweight. If the contract provides for more than one aircraft, the overweight or underweight shall be taken as the average of the overweight or underweight values for the aircraft as finally accepted and shown in the required actual weight data.

b. Production contracts: For production contracts, the overweight or underweight shall be taken as the average of the overweight and underweight values shown by the required actual weight data. Refer to paragraphs 3.7.4, 3.7.5 and 3.7.8.

3.11.3 Engine.

a. Dry weight: The items included in the dry weight of the engine shall be those defined in the applicable engine model specification. The engine dry weight entered in the engine log by the engine manufacturer may be used as a basis for determining the "as installed" weight for all but the Actual Weight Report - First Aircraft and the first aircraft equipped with an engine having a new model designation.

b. Estimated weight: The estimated "as installed" weight of engines shall be the specified dry weight plus an allowance as follows for the oil and grease contained in the engine:

Reciprocating	15 pounds
Turbopropeller	10 pounds
Turbojet	5 pounds

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c. Actual weight: The actual "as installed" engine weight shall be that weight obtained when the engine, including all dry weight items, is cleaned up and ready for installation in the aircraft.

4. QUALITY ASSURANCE PROVISIONS

4.1 Methods of inspection and approval.

4.2 General. The Government Inspector shall certify to the correctness of all actual weight data submitted to the Government and shall witness actual weighings and sign the official weighing form. The Inspector shall also comment in appropriate detail on all weight estimates and discussions submitted in correspondence concerning proposed contract changes or amendments.

4.3 Final approval. Final approval shall be obtained from the activity under which procurement is made.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Weight and balance control data shall be packaged in such manner that it will not be damaged in transit by common carrier.

5.2 Delivery. Shall be as directed by the procuring activity.

5.3 Marking of shipments. There are no requirements for the marking of the packages containing weight and balance control data.

6. NOTES AND CONCLUDING MATERIAL

6.1 Intended use. The specification is used in the management of Weight and Balance Control Systems and in determining compliance with contractual requirements during design and construction of aircraft.

6.2 Ordering data. Procurement Documents should specify the following:

a. Title, Number, and Date of this specification.

6.2.1 Procurement requirements. Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

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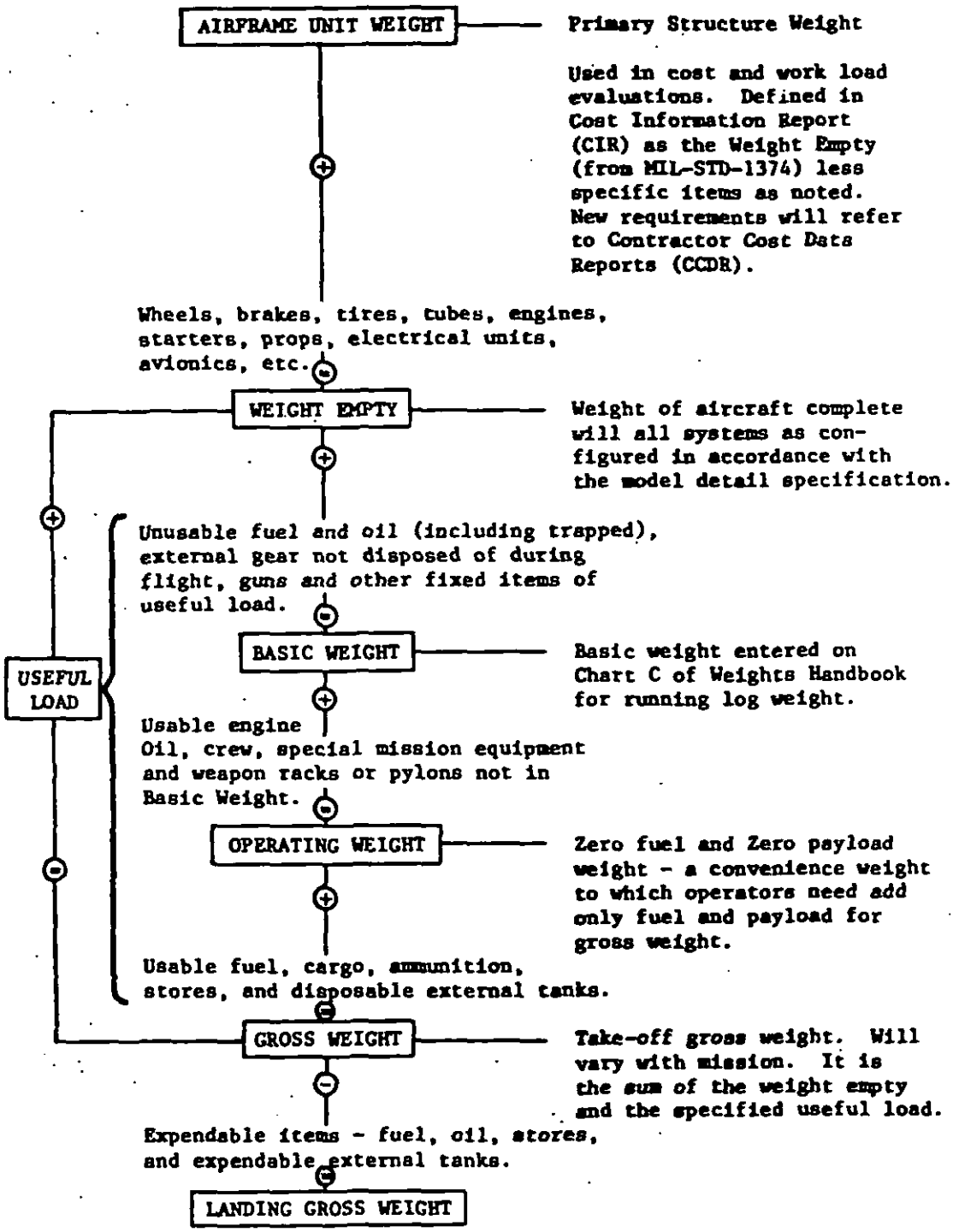
6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID		
		Air Force	Army	NAVAIR
3.2	Weight Control Management Program Plan Report	DI-S-3572/ S-149-1	DI-E-1124	
3.3.1	Specification Weight Derivation			UDI-S-21189
3.3.2	Post-Design Weight Analysis	DI-S-3584/ S-103-2	DI-E-1124	UDI-S-21188
3.7.1	Weight & Balance Status	DI-S-3584/ S-103-2	DI-E-1124	UDI-S-21184
3.7.2	Estimated Weight Report	DI-S-3584/ S-103-2	DI-E-1124	UDI-S-21183
3.7.3	Calculated Weight Report	DI-S-3584/ S-103-2	DI-E-1124	UDI-S-21185
3.7.4	Actual Weight Report (Detail)	DI-S-3584/ S-103-2	DI-E-1124	UDI-S-21187
3.7.5	Actual Weight Report (Intermediate)	DI-S-3584/ S-103-2	DI-E-1124	UDI-S-21181A
3.7.6 & 3.7.7	Appendix to Actual Weight Reports	DI-S-3584/ S-103-2	DI-E-1124	UDI-S-21187
3.7.8	Actual Weight Report (Last)	DI-S-3584/ S-103-2	DI-E-1124	UDI-S-21187
3.7.9	Handbook of Weight and Balance	DI-S-3584/ S-103-2	DI-E-1124	UDI-M-21186
3.7.10	Balance Computer Design Data	DI-S-3584/ S-103-2	DI-E-1124	UDI-M-21186
3.6.3	Engineering Change Proposal (ECP)		DI-E-1124	
3.6.5	Aircraft Modification - Prior to Service	DI-S-3584/ S-103-2	DI-E-1124	
3.6.6	Aircraft Modification - Return from Service	DI-S-3584/ S-103-2	DI-E-1124	

Such data will be delivered as described on approved (numbered) DIDs (Data Item Description DD Form 1664) when specified on DD Form 1423 (Contract Data Requirements List) and incorporated into the applicable contract.

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6.2.3 Weight definitions. As used in MIL-STD-1374 and Weight Handbooks.



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6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Navy - AS
Air Force - 11
Army - AV

Preparing Activity:

Navy - AS

Project No. 1500-0093

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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